INCH-POUND

MIL-M-28787/362
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

DRIVER/RECEIVER DIFFERENTIAL MODULE, KEY CODE HJN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic driver/receiver differential module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 3.5 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be HJN.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 100 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B	Vcc stress	III	S	---------Module shall pass all remaining tests---------						
C1-C28	Ibd	IV	S	0	100	0	100	0	100	uA
C29-C32	Ibd	IV	S	0	200	0	200	0	200	uA
D1-D16	Ios	V	S	-120	-40	-120	-40	-120	-40	mA
D17-D24	Ios	V	S	-80	-15	-80	-15	-80	-15	mA
E1-E16	Iozl	VI	X	-80	80	-80	80	-80	80	uA
F1-F16	Iozh	VII	X	-80	80	-80	80	-80	80	uA
F17-F24	Iozh	VII	X	0	200	0	200	0	200	uA
G1-G32	Vic	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H24	Vol	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J16	Voh	X	X	2.0	4.5	2.0	4.5	2.0	4.5	V
J17-J24	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K8	Iil	XI	X	-0.7	-1.6	-0.7	-1.6	-0.7	-1.6	mA
K9-K28	Iil	XI	X	-30.0	-400.0	-30.0	-400.0	-30.0	-400.0	uA
K29-K32	Iil	XI	X	-60.0	-800.0	-60.0	-800.0	-60.0	-800.0	uA
K33-K48	Iil	XI	X	-0.2	-0.9	-0.2	-0.9	-0.2	-0.9	mA
K49-K52	Iil	XI	X	-10.17	-11.47	-10.17	-11.47	-10.17	-11.47	mA
L1-L8	Iih	XII	X	0	40	0	40	0	40	uA
L9-L28	Iih	XII	X	0	20	0	20	0	20	uA
L29-L32	Iih	XII	X	0	40	0	40	0	40	uA
M1	Icc	XIII	X	21.1	211.0	21.1	211.0	21.1	211.0	mA
M2	Icc	XIII	X	28.2	282.0	28.2	282.0	28.2	282.0	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
N1,N2	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N3	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N4,N5	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N6	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N7,N8	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N9	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N10,N11	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N12	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N13,N14	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N15	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N16,N17	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N18	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N19,N20	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
N21	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N22,N23	Propagation delay time (tPHL)	XIV	S	5	52	5	52	5	52	nS
N24	Propagation delay time (tPHL)	XIV	S	5	82	5	82	5	82	nS
N25,N26	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N27	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N28,N29	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N30	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N31,N32	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N33	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N34,N35	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N36	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N37,N38	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N39	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
N40,N41	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N42	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N43,N44	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N45	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N46,N47	Propagation delay time (tPHL)	XIV	S	5	81	5	81	5	81	nS
N48	Propagation delay time (tPHL)	XIV	S	5	58	5	58	5	58	nS
N49-N52	Propagation delay time (tPHL)	XIV	S	4	73	4	73	4	73	nS
N53-N56	Propagation delay time (tPHL)	XIV	S	4	82	4	82	4	82	nS
N57-N64	Propagation delay time (tPHL)	XIV	S	2	58	2	58	2	58	nS
P1,P2	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P3	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P4,P5	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P6	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P7, P8	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P9	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P10, P11	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P12	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P13, P14	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P15	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P16, P17	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P18	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P19, P20	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P21	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS
P22, P23	Propagation delay time (tPLH)	XV	S	5	55	5	55	5	55	nS
P24	Propagation delay time (tPLH)	XV	S	5	85	5	85	5	85	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P25,P26	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P27	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P28,P29	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P30	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P31,P32	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P33	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P34,P35	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P36	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P37,P38	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P39	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P40,P41	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P42	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P43,P44	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS

- See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P45	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P46,P47	Propagation delay time (tPLH)	XV	S	5	92	5	92	5	92	nS
P48	Propagation delay time (tPLH)	XV	S	5	72	5	72	5	72	nS
P49-P52	Propagation delay time (tPLH)	XV	S	4	85	4	85	4	85	nS
P53-P56	Propagation delay time (tPLH)	XV	S	4	76	4	76	4	76	nS
P57-P64	Propagation delay time (tPLH)	XV	S	2	72	2	72	2	72	nS
Q1-Q16	Output enable delay time (tPZH)	XVI	S	2	27	2	27	2	27	nS
R1-R16	Output disable delay time (tPHZ)	XVII	Q	2	25	2	25	2	25	nS
S1-S16	Output enable delay time (tPZL)	XVIII	S	2	38	2	38	2	38	nS
T1-T16	Output disable delay time (tPLZ)	XIX	Q	2	47	2	47	2	47	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V	Functional	XX	X	----------		See table XX				
W1-W3	Continuity	XXI	X	0	1.0	0	1.0	0	1.0	Ohms
W4-W6	Continuity	XXI	Q	0	0.1	0	0.1	0	0.1	Ohms
X1-X16	Output crosstalk (low level)	XXII	Q	0	400	0	400	0	400	mV
Y1-Y16	Output crosstalk (high level)	XXIII	Q	0	400	0	400	0	400	mV
Z	PSTA	XXIV	Q	0	500	0	500	0	500	mV
AA1, AA2	PSFC	XXV	S	3.6	6.6	3.6	6.6	3.6	6.6	uF
BB1-BB48	Cin	XXVI	Q	2	15	2	15	2	15	pF
CC1-CC8	Terminator	XXVII	S	0.394	0.591	0.394	0.591	0.394	0.591	V
CC9-CC16	Terminator	XXVII	S	2.79	4.18	2.79	4.18	2.79	4.18	V

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/362-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

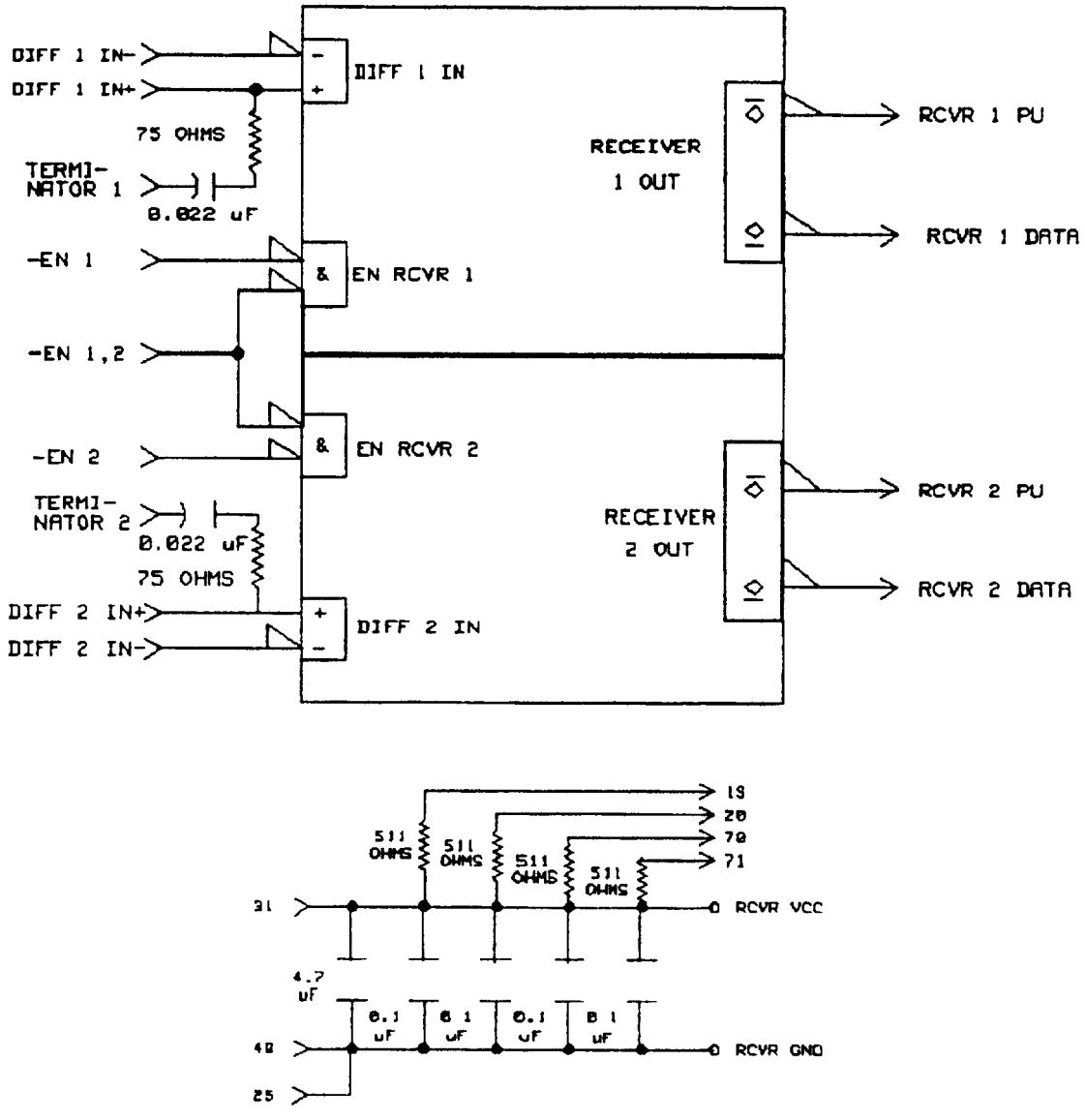
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

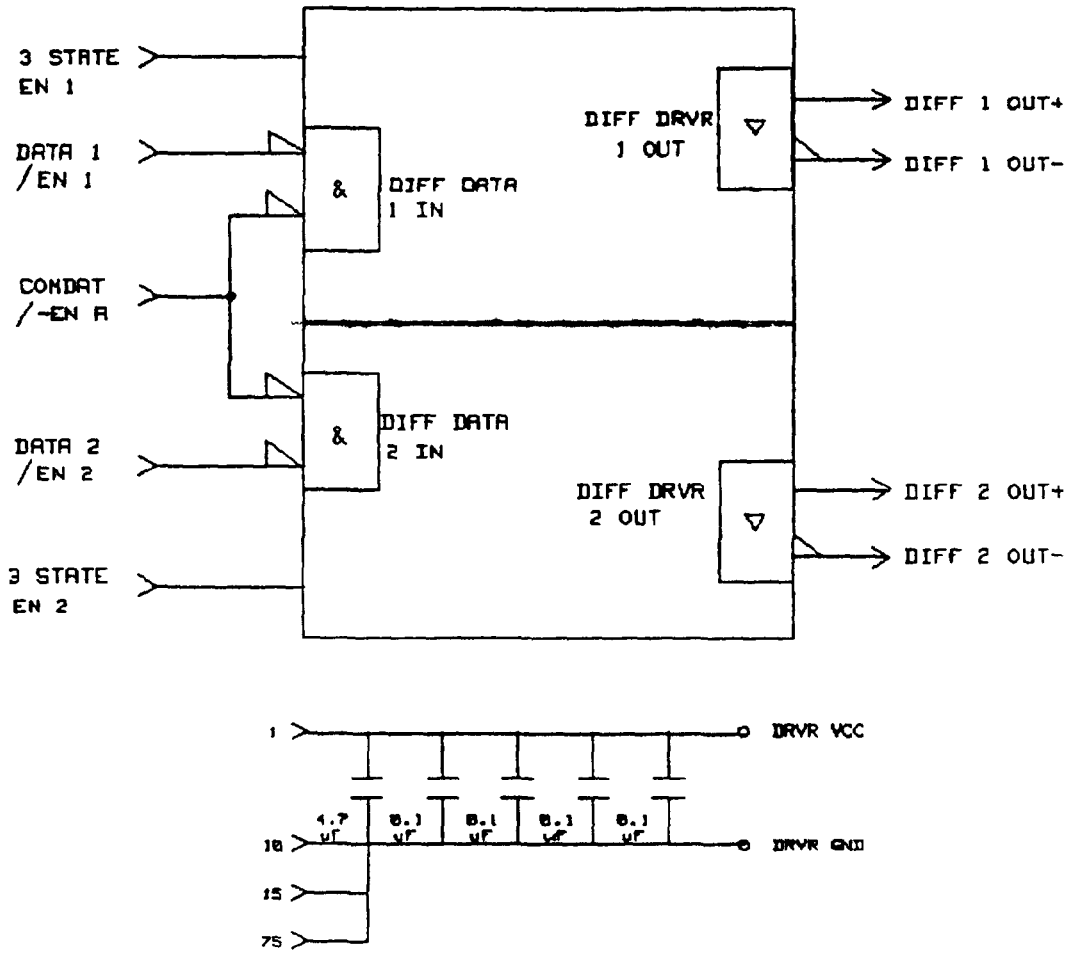


NOTE:

1. The functional block diagram is one of four identical differential receivers on the module.

FIGURE 1a. HJN functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTE:

1. The functional block diagram is one of four identical differential receivers on the module.

FIGURE 1b. HJN functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

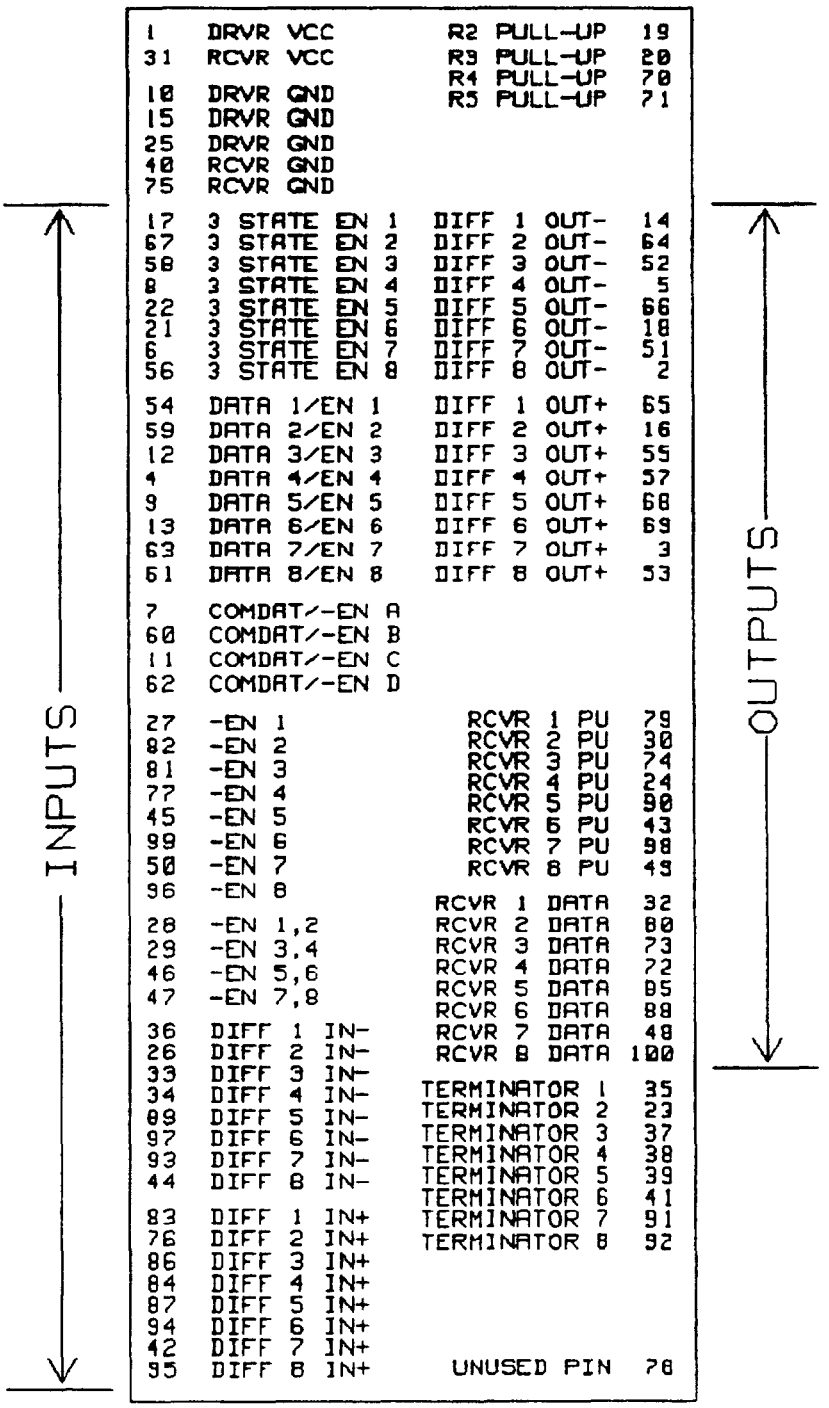


FIGURE 2. HJN pin assignment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A, B, C, D and G are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D and G where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E, F, H through AA and CC shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

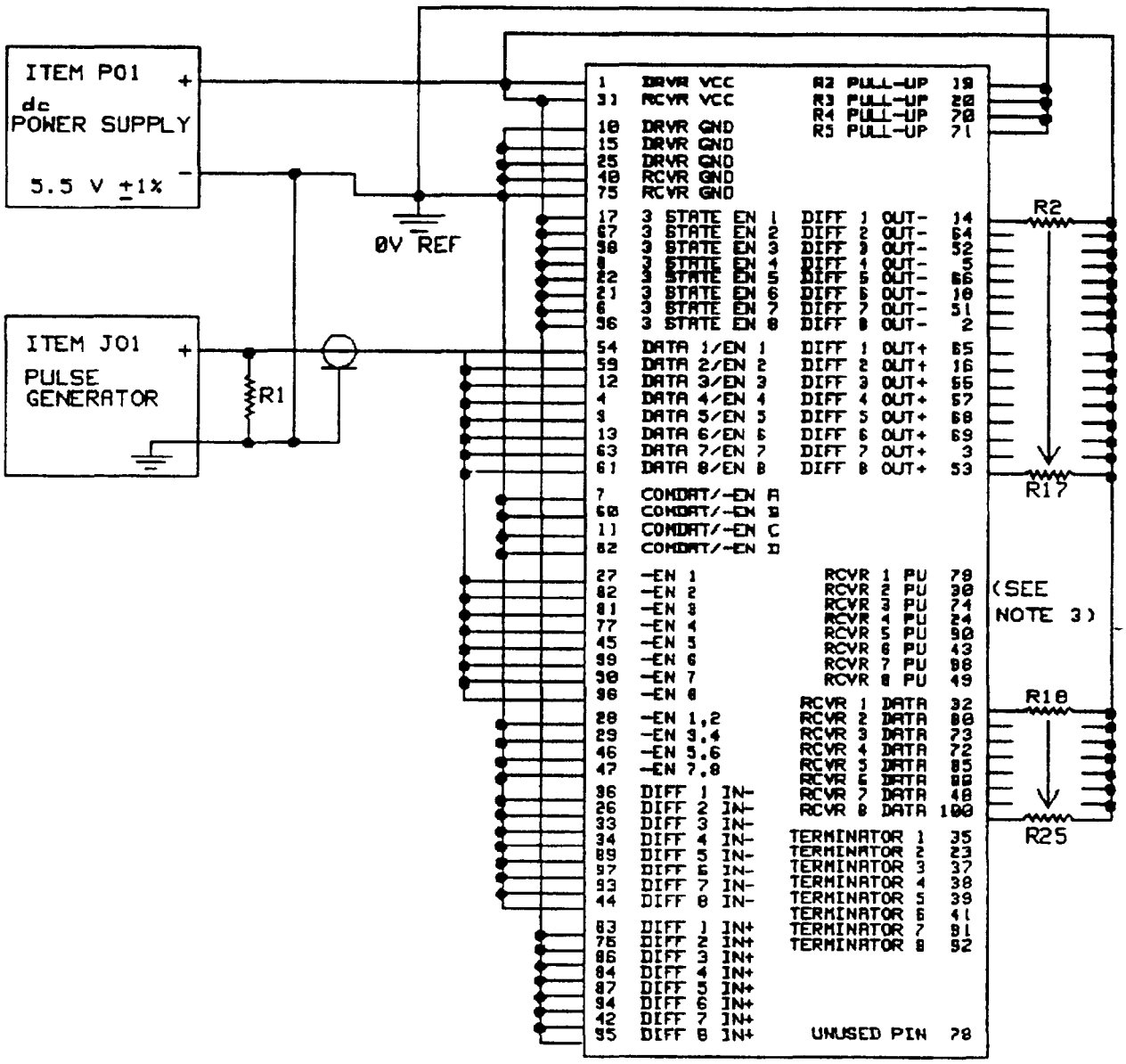


FIGURE 3. HJN operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

NOTES:

1. Item JO1 setup: PA = 3.0 ± 0.1 V, PRR = 1 MHz ± 20 percent, PW = 500 ± 100 nS, tTLH = tTHL = 10 ± 2 nS.

2. Resistor values: R1 = 51.1 Ohms ± 5 percent, 1 W; R2-R17 = 127 Ohms ± 1 percent, 1/4 W; R18-R25 = 348 Ohms ± 1 percent, 1/8 W.

3. Connect the RCVR x PU pin to the respective RCVR x DATA pin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

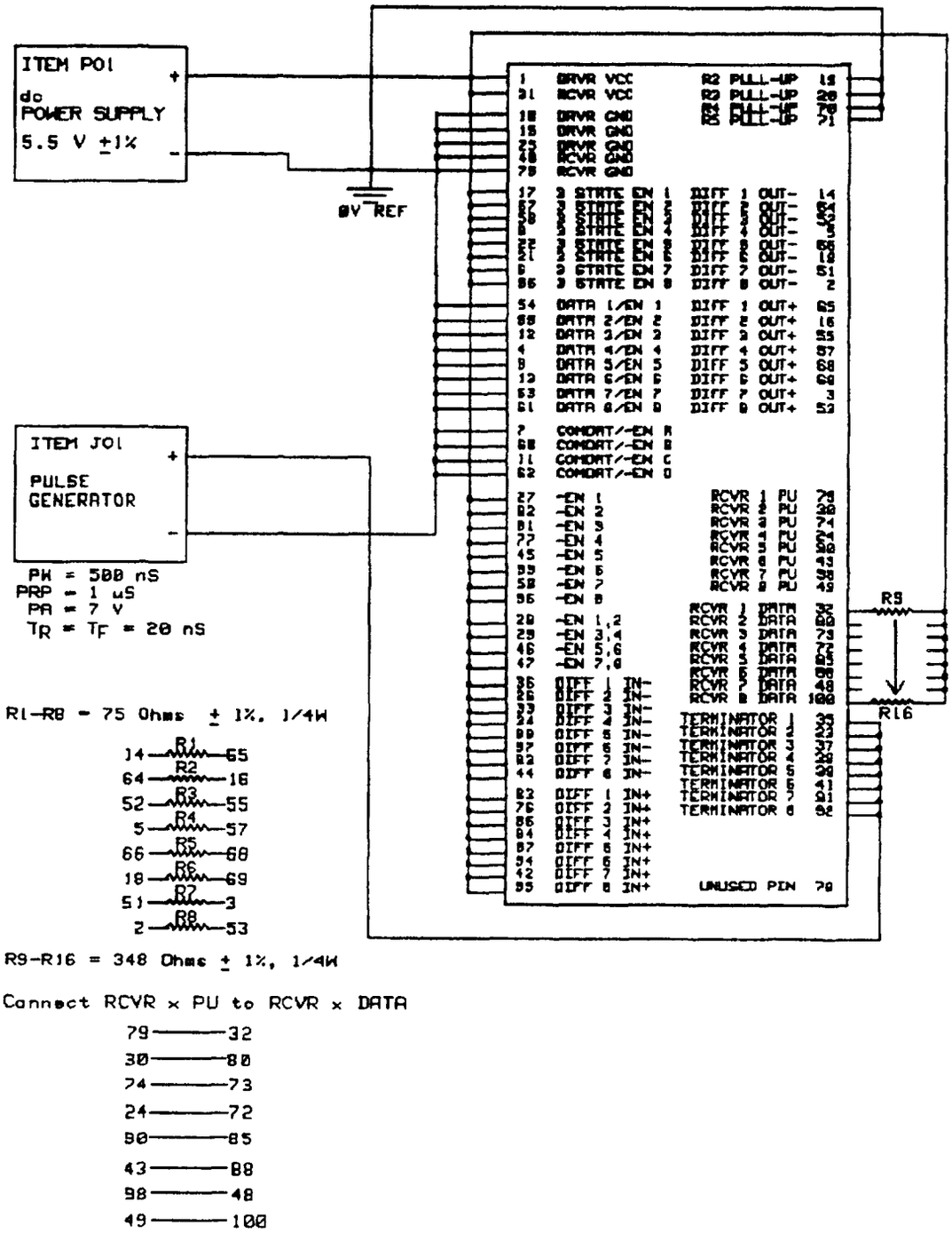


FIGURE 4. HJN component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

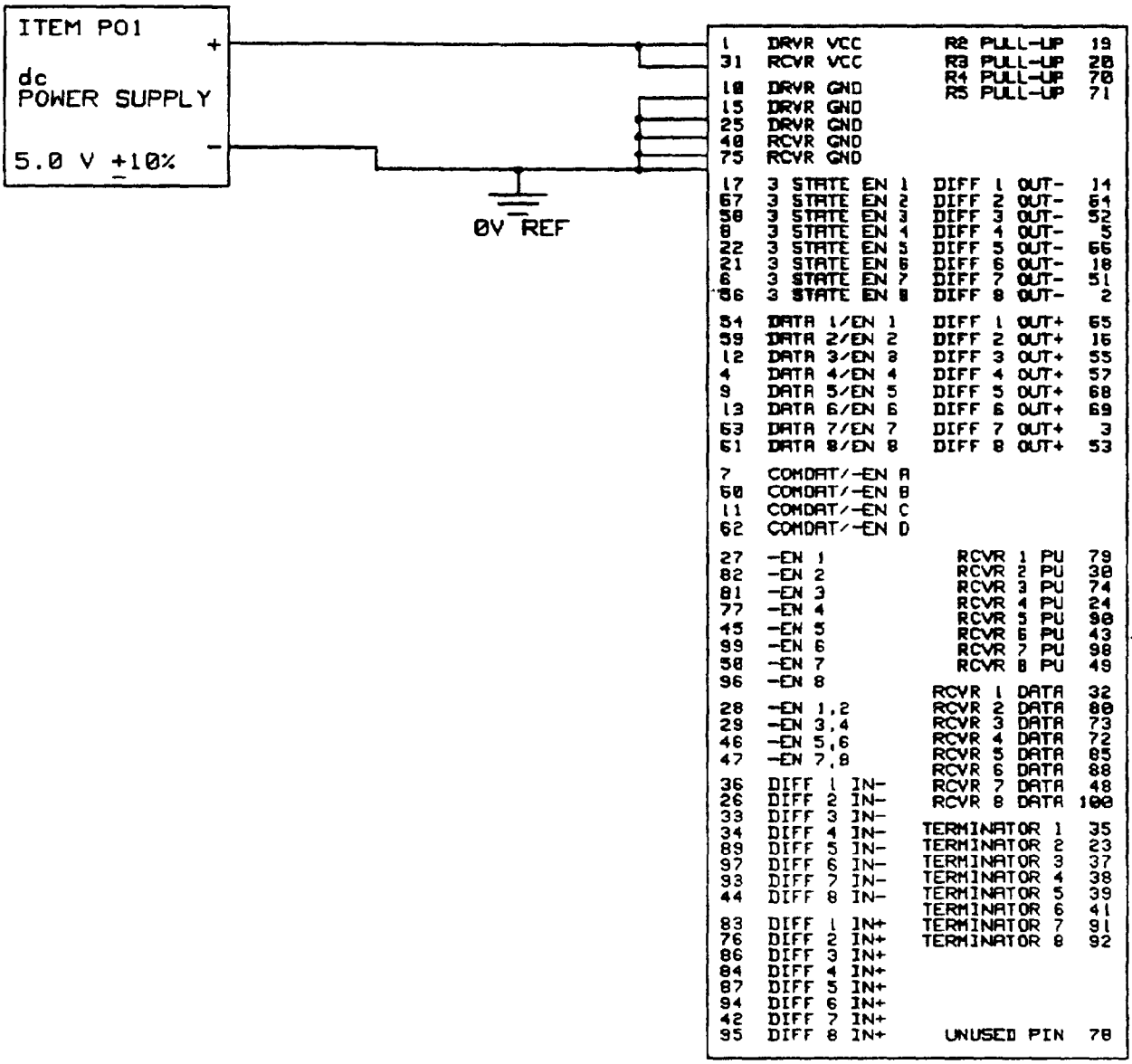


FIGURE 5. HJN vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E, F, H through AA and CC shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests E, F, H through AA and CC shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 2.77 W (max); 2.52 W (typical).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil max (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 4/-EN 4	-1.5	0.7	2	-400	20	100	15
COMDAT/-EN A	-1.5	0.7	2	-400	20	100	15
DATA 5/-EN 5	-1.5	0.7	2	-400	20	100	15
COMDAT/-EN C	-1.5	0.7	2	-400	20	100	15
DATA 2/-EN 2	-1.5	0.7	2	-400	20	100	15
DATA 6/-EN 6	-1.5	0.7	2	-400	20	100	15
-EN 1	-1.5	0.7	2	-400	20	100	15
-EN 1,2	-1.5	0.7	2	-800	40	200	15
-EN 3,4	-1.5	0.7	2	-800	40	200	15
-EN 5	-1.5	0.7	2	-400	20	100	15
-EN 5,6	-1.5	0.7	2	-800	40	200	15
-EN 7,8	-1.5	0.7	2	-800	40	200	15
-EN 7	-1.5	0.7	2	-400	20	100	15
DATA 1/-EN 1	-1.5	0.7	2	-400	20	100	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil max (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 2/-EN 2	-1.5	0.7	2	-400	20	100	15
COMDAT/-EN B	-1.5	0.7	2	-400	20	100	15
DATA 8/-EN 8	-1.5	0.7	2	-400	20	100	15
COMDAT/-EN D	-1.5	0.7	2	-400	20	100	15
DATA 7/-EN 7	-1.5	0.7	2	-400	20	100	15
-EN 4	-1.5	0.7	2	-400	20	100	15
-EN 3	-1.5	0.7	2	-400	20	100	15
-EN 2	-1.5	0.7	2	-400	20	100	15
-EN 8	-1.5	0.7	2	-400	20	100	15
-EN 6	-1.5	0.7	2	-400	20	100	15
3 STATE EN 7	-1.5	0.8	2	-1600	40	100	15
3 STATE EN 4	-1.5	0.8	2	-1600	40	100	15
3 STATE EN 1	-1.5	0.8	2	-1600	40	100	15
3 STATE EN 6	-1.5	0.8	2	-1600	40	100	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil max (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
3 STATE EN 5	-1.5	0.8	2.0	-1600	40	100	15
3 STATE EN 8	-1.5	0.8	2.0	-1600	40	100	15
3 STATE EN 3	-1.5	0.8	2.0	-1600	40	100	15
3 STATE EN 2	-1.5	0.8	2.0	-1600	40	100	15
DIFF 2 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 3 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 4 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 1 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 7 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 8 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 2 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 1 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 4 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 3 IN+	NA	-0.5	0.5	-900	NA	NA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil max (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DIFF 5 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 5 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 7 IN-	NA	-0.5	0.5	-900	NA	NA	15
DIFF 6 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 8 IN+	NA	-0.5	0.5	-900	NA	NA	15
DIFF 6 IN-	NA	-0.5	0.5	-900	NA	NA	15

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)
DIFF 8 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 7 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 4 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 1 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 2 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 6 OUT-	2.0	0.4	-40	40	80	-80	-40

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)
DIFF 7 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 3 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 8 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 3 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 4 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 2 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 1 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 5 OUT-	2.0	0.4	-40	40	80	-80	-40
DIFF 5 OUT+	2.0	0.4	-40	40	80	-80	-40
DIFF 6 OUT+	2.0	0.4	-40	40	80	-80	-40
RCVR 1 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 1 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 7 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 7 PU	2.4	0.4	-5	16	NA	NA	-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)
RCVR 4 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 4 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 3 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 3 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 2 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 2 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 5 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 5 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 6 DATA	2.4	NA	-5	16	200	NA	NA
RCVR 6 PU	2.4	0.4	-5	16	NA	NA	-15
RCVR 8 DATA	2.4	NA	-5	26	200	NA	NA
RCVR 8 PU	2.4	0.4	-5	16	NA	NA	-15

6.3.2.1 Signal pairs. The PU and DATA signals are pull-up and open collector pairs. The Voh and Vol parameters are valid only when the pairs are tested together.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.4 Module description. The module contains eight DC differential line drivers and eight differential line receivers. The line drivers are designed to drive a 75 Ohm cable and are three statable. The line receivers have a pin programmable active pull-up. They can be operated as a normal totem pole output or as an open collector device. There are four 510 Ohm pull-up resistors and eight AC coupled Ohm Terminators.

6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
3 STATE EN 1	17	I	Enable control liner for the corresponding differential line driver outputs. A high level enables the differential line drive, a low level places the differential line driver in the high ımpedance state.
3 STATE EN 2	67	I	
3 STATE EN 3	58	I	
3 STATE EN 4	8	I	
3 STATE EN 5	22	I	
3 STATE EN 6	21	I	
3 STATE EN 7	6	I	
3 STATE EN 8	56	I	
DATA 1/EN 1	54	I	Data input for the differential line drivers COMDAT/-EN A-COMDAT/-EN D must be low to enable data input.
DATA 2/EN 2	59	I	
DATA 3/EN 3	12	I	
DATA 4/EN 4	4	I	
DATA 5/EN 5	9	I	
DATA 6/EN 6	13	I	
DATA 7/EN 7	63	I	
DATA 8/EN 8	61	I	
COM DAT/-EN A	7	I	Data input which enables both of the dual line drivers. COMDAT/-EN A enables DATA 1/EN 1 and DATA 2/EN 2-- COMDAT/-END enables DATA 7/EN 7 and DATA 8/EN 8.
COM DAT/-EN B	60	I	
COM DAT/-EN C	11	I	
COM DAT/-EN D	62	I	
DIFF 1 OUT-	14	O	NAND output of the differential line drivers.
DIFF 2 OUT-	64	O	
DIFF 3 OUT-	62	O	
DIFF 4 OUT-	5	O	
DIFF 5 OUT-	66	O	
DIFF 6 OUT-	18	O	
DIFF 7 OUT-	51	O	
DIFF 8 OUT-	2	O	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
DIFF 1 OUT+	65	O	AND output of the differential line drivers.
DIFF 2 OUT+	16	O	
DIFF 3 OUT+	55	O	
DIFF 4 OUT+	57	O	
DIFF 5 OUT+	68	O	
DIFF 6 OUT+	69	O	
DIFF 7 OUT+	3	O	
DIFF 8 OUT+	53	O	
-EN 1	27	I	Active low control line that enables the corresponding differential line receiver input.
-EN 2	82	I	
-EN 3	81	I	
-EN 4	77	I	
-EN 5	45	I	
-EN 6	99	I	
-EN 7	50	I	
-EN 8	96	I	
-EN 1,2	28	I	Active low control line that, in conjunction with -EN 1 through -EN 8, enables the corresponding differential line receiver inputs.
-EN 3,4	29	I	
-EN 5,6	46	I	
-EN 7,8	47	I	
DIFF 1 IN-	36	I	Input to the negative differential input terminal.
DIFF 2 IN-	26	I	
DIFF 3 IN-	33	I	
DIFF 4 IN-	34	I	
DIFF 5 IN-	89	I	
DIFF 6 IN-	97	I	
DIFF 7 IN-	93	I	
DIFF 8 IN-	44	I	
DIFF 1 IN+	83	I	Input to the positive differential input terminal.
DIFF 2 IN+	76	I	
DIFF 3 IN+	86	I	
DIFF 4 IN+	84	I	
DIFF 5 IN+	87	I	
DIFF 6 IN+	94	I	
DIFF 7 IN+	42	I	
DIFF 8 IN+	95	I	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	
TERMINATOR 1	35	I	75 Ohm AC impedance terminators for use with the differential line receivers.
TERMINATOR 2	23	I	
TERMINATOR 3	37	I	
TERMINATOR 4	38	I	
TERMINATOR 5	39	I	
TERMINATOR 6	41	I	
TERMINATOR 7	91	I	
TERMINATOR 8	92	I	
RCVR 1 PU	79	O	Active pull up output of the differential line receiver.
RCVR 2 PU	30	O	
RCVR 3 PU	74	O	
RCVR 4 PU	24	O	
RCVR 5 PU	90	O	
RCVR 6 PU	43	O	
RCVR 7 PU	98	O	
RCVR 8 PU	49	O	
RCVR 1 DATA	32	O	Active pull down output of the differential line receiver.
RCVR 2 DATA	80	O	
RCVR 3 DATA	73	O	
RCVR 4 DATA	72	O	
RCVR 5 DATA	85	O	
RCVR 6 DATA	88	O	
RCVR 7 DATA	48	O	
RCVR 8 DATA	100	O	
R2 PULL-UP	19	O	510 Ohm pull up resistor for use with RCVR 1 DATA-RCVR 8 DATA outputs to implement the AND function.
R3 PULL-UP	20	O	
R4 PULL-UP	70	O	
R5 PULL-UP	71	O	

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 6.

(b) Vcc stress test - table III, figure 7.

(c) Ibd test - table IV, figure 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

(d) Ios test - table V, figure 9.

(e) Iozl test - table VI, figure 10.

(f) Iozh test - table VII, figure 10.

(g) Vic test - table VIII, figure 11.

(h) Vol test - table IX, figures 12 and 13.

(i) Voh test - table X, figures 12 and 13.

(j) Iil test - table XI, figure 14.

(k) Iih test - table XII, figure 14.

(l) Icc test - table XIII, figure 15.

(m) Propagation delay time test - tables XIV and XV, figures 16 and 17.

(n) Output enable delay time test - tables XVI and XVIII, figures 18 and 19.

(o) Output disable delay time test - tables XVII and XIX, figures 18 and 19.

(p) Functional test - table XX, figure 20.

(q) Continuity test - table XXI, figure 21.

(r) Output crosstalk tests - tables XXII and XXIII, figures 22 and 23.

(s) PSTA test - table XXIV, figures 24 and 25.

(t) PSFC test - table XXV, figure 26.

(u) Cin test - table XXVI, figure 26.

(v) Terminator test - table XXVII, figures 27 and 28.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0058)

Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	6
A2	Isolation, unused to used pins	78	Remaining pins	10	INF	MOhm	6

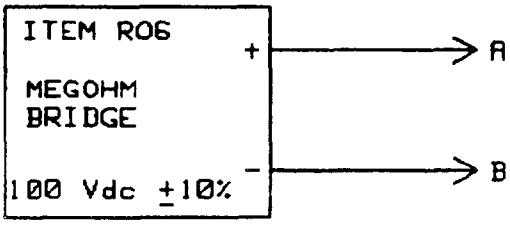
NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

3. The minimum end-of-life limit is 1 MOhm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



1	DRVR VCC	R2 PULL-UP	19
31	RCVR VCC	R3 PULL-UP	28
		R4 PULL-UP	70
10	DRVR GND	R5 PULL-UP	71
15	DRVR GND		
25	DRVR GND		
40	RCVR GND		
75	RCVR GND		
17	3 STATE EN 1	DIFF 1 OUT-	14
67	3 STATE EN 2	DIFF 2 OUT-	64
58	3 STATE EN 3	DIFF 3 OUT-	52
8	3 STATE EN 4	DIFF 4 OUT-	5
22	3 STATE EN 5	DIFF 5 OUT-	66
21	3 STATE EN 6	DIFF 6 OUT-	18
6	3 STATE EN 7	DIFF 7 OUT-	51
56	3 STATE EN 8	DIFF 8 OUT-	2
54	DATA 1/EN 1	DIFF 1 OUT+	65
59	DATA 2/EN 2	DIFF 2 OUT+	16
12	DATA 3/EN 3	DIFF 3 OUT+	55
4	DATA 4/EN 4	DIFF 4 OUT+	57
9	DATA 5/EN 5	DIFF 5 OUT+	68
13	DATA 6/EN 6	DIFF 6 OUT+	69
63	DATA 7/EN 7	DIFF 7 OUT+	3
61	DATA 8/EN 8	DIFF 8 OUT+	53
7	COMDAT/-EN A		
60	COMDAT/-EN B		
11	COMDAT/-EN C		
62	COMDAT/-EN D		
27	-EN 1	RCVR 1 PU	79
82	-EN 2	RCVR 2 PU	30
81	-EN 3	RCVR 3 PU	74
77	-EN 4	RCVR 4 PU	24
45	-EN 5	RCVR 5 PU	90
99	-EN 6	RCVR 6 PU	43
50	-EN 7	RCVR 7 PU	98
96	-EN 8	RCVR 8 PU	49
28	-EN 1,2	RCVR 1 DATA	32
29	-EN 3,4	RCVR 2 DATA	80
46	-EN 5,6	RCVR 3 DATA	73
47	-EN 7,8	RCVR 4 DATA	72
		RCVR 5 DATA	85
36	DIFF 1 IN-	RCVR 6 DATA	88
26	DIFF 2 IN-	RCVR 7 DATA	48
33	DIFF 3 IN-	RCVR 8 DATA	100
34	DIFF 4 IN-	TERMINATOR 1	35
89	DIFF 5 IN-	TERMINATOR 2	23
97	DIFF 6 IN-	TERMINATOR 3	37
93	DIFF 7 IN-	TERMINATOR 4	38
44	DIFF 8 IN-	TERMINATOR 5	39
		TERMINATOR 6	41
83	DIFF 1 IN+	TERMINATOR 7	91
76	DIFF 2 IN+	TERMINATOR 8	92
86	DIFF 3 IN+		
84	DIFF 4 IN+		
87	DIFF 5 IN+		
94	DIFF 6 IN+		
42	DIFF 7 IN+		
95	DIFF 8 IN+	UNUSED PIN	78

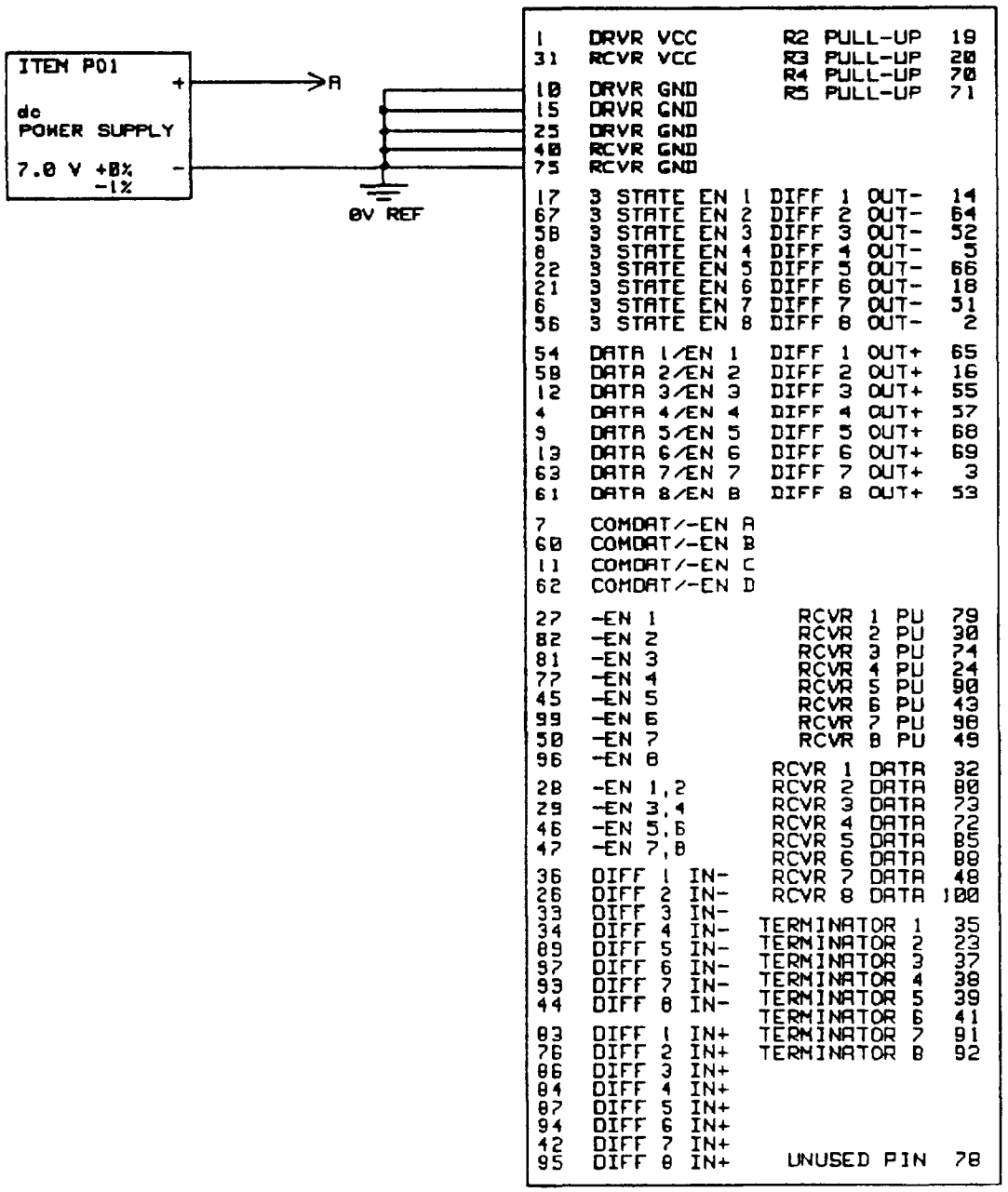
FIGURE 6. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Fig
			Positive lead to	Negative lead to		
B	Vcc	7.0	1,31	10,15, 25,40, 75	Module shall pass all remaining tests.	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTE:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

FIGURE 7. Vcc stress test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test.

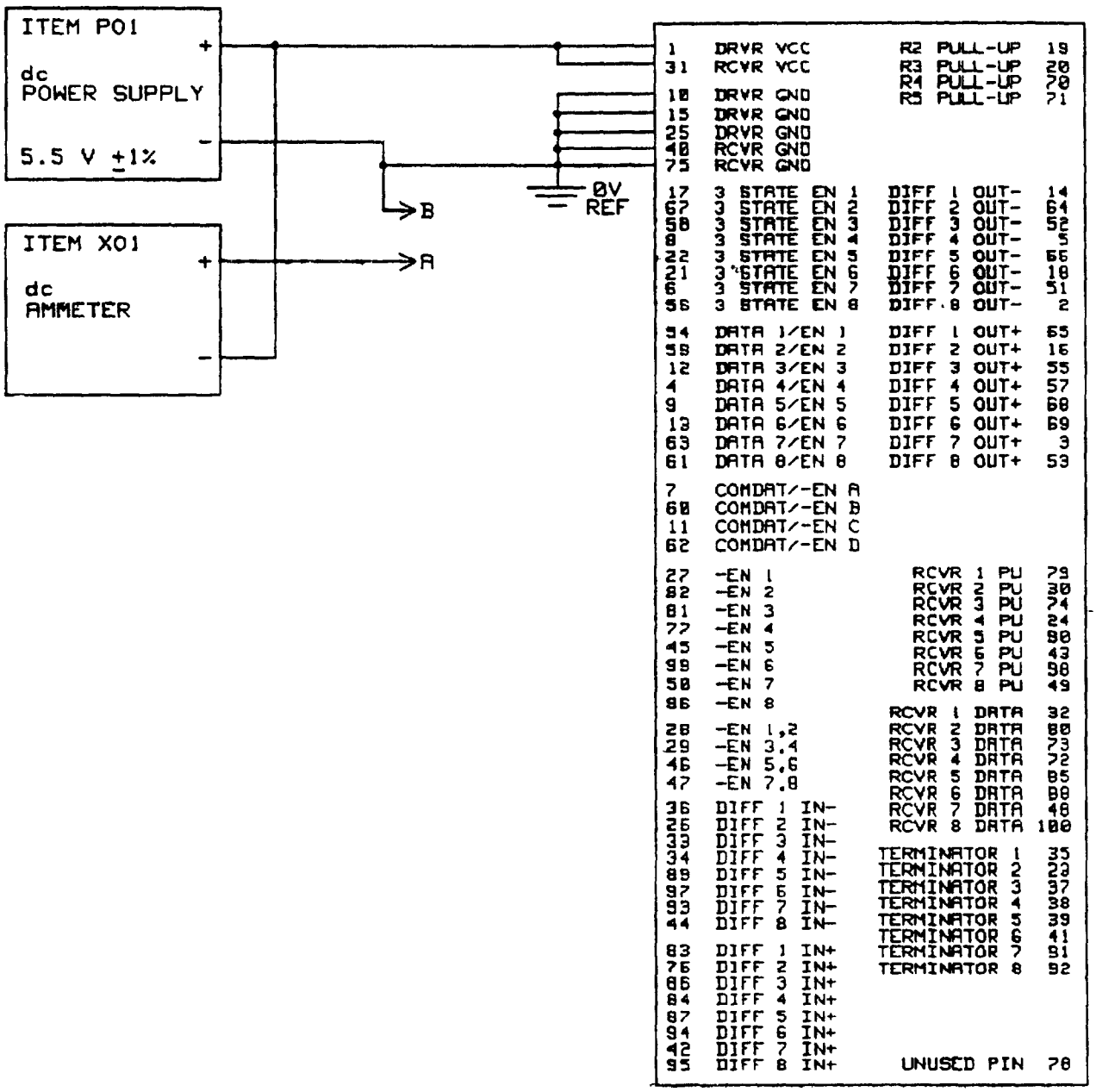
Test cond	Signal name	A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
C1	3 STATE EN 1	17	0	100	uA	8
C2	3 STATE EN 2	67				
C3	3 STATE EN 3	58				
C4	3 STATE EN 4	8				
C5	3 STATE EN 5	22				
C6	3 STATE EN 6	21				
C7	3 STATE EN 7	6				
C8	3 STATE EN 8	56				
C9	DATA 1/EN 1	54				
C10	DATA 2/EN 2	59				
C11	DATA 3/EN 3	12				
C12	DATA 4/EN 4	4				
C13	DATA 5/EN 5	9				
C14	DATA 6/EN 6	13				
C15	DATA 7/EN 7	63				
C16	DATA 8/EN 8	61				
C17	COMDAT/-EN A	7				
C18	COMDAT/-EN B	60				
C19	COMDAT/-EN C	11				
C20	COMDAT/-EN D	62	0	100	uA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test - Continued.

Test cond	Signal name	A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
C21	-EN 1	27	0	100	uA	8
C22	-EN 2	82				
C23	-EN 3	81				
C24	-EN 4	77				
C25	-EN 5	45				
C26	-EN 6	99				
C27	-EN 7	50				
C28	-EN 8	96	0	100	uA	8
C29	-EN 1,2	28	0	200	uA	8
C30	-EN 3,4	29				
C31	-EN 5,6	46				
C32	-EN 7,8	47	0	200	uA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTE:

1. All input pins shall be connected to B, except the pin connected to A.

FIGURE 8. Input breakdown current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE V. Output short circuit current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial T limits Min	25 ± 5°C Initial T limits Max	Units	Fig
D1	DIFF 1 OUT-	14	17,54	Remaining input pins	-120	-40	mA	9
D2	DIFF 2 OUT-	64	59,67					
D3	DIFF 3 OUT-	52	12,58					
D4	DIFF 4 OUT-	5	4,8					
D5	DIFF 5 OUT-	66	9,22					
D6	DIFF 6 OUT-	18	13,21					
D7	DIFF 7 OUT-	51	6,63					
D8	DIFF 8 OUT-	2	56,61					
D9	DIFF 1 OUT+	65	17					
D10	DIFF 2 OUT+	16	67					
D11	DIFF 3 OUT+	55	58					
D12	DIFF 4 OUT+	57	8					
D13	DIFF 5 OUT+	68	22					
D14	DIFF 6 OUT+	69	21					
D15	DIFF 7 OUT+	3	6	Remaining input pins	-120	-40	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE V. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
D16	DIFF B OUT+	53	56	Remaining input pins	-120	-40	mA	9
D17	RCVR 1 PU	79	36		-80	-15	mA	9
D18	RCVR 2 PU	30	26					
D19	RCVR 3 PU	74	33					
D20	RCVR 4 PU	24	34					
D21	RCVR 5 PU	90	89					
D22	RCVR 6 PU	43	97					
D23	RCVR 7 PU	98	93					
D24	RCVR 8 PU	49	44	Remaining input pins	-80	-15	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

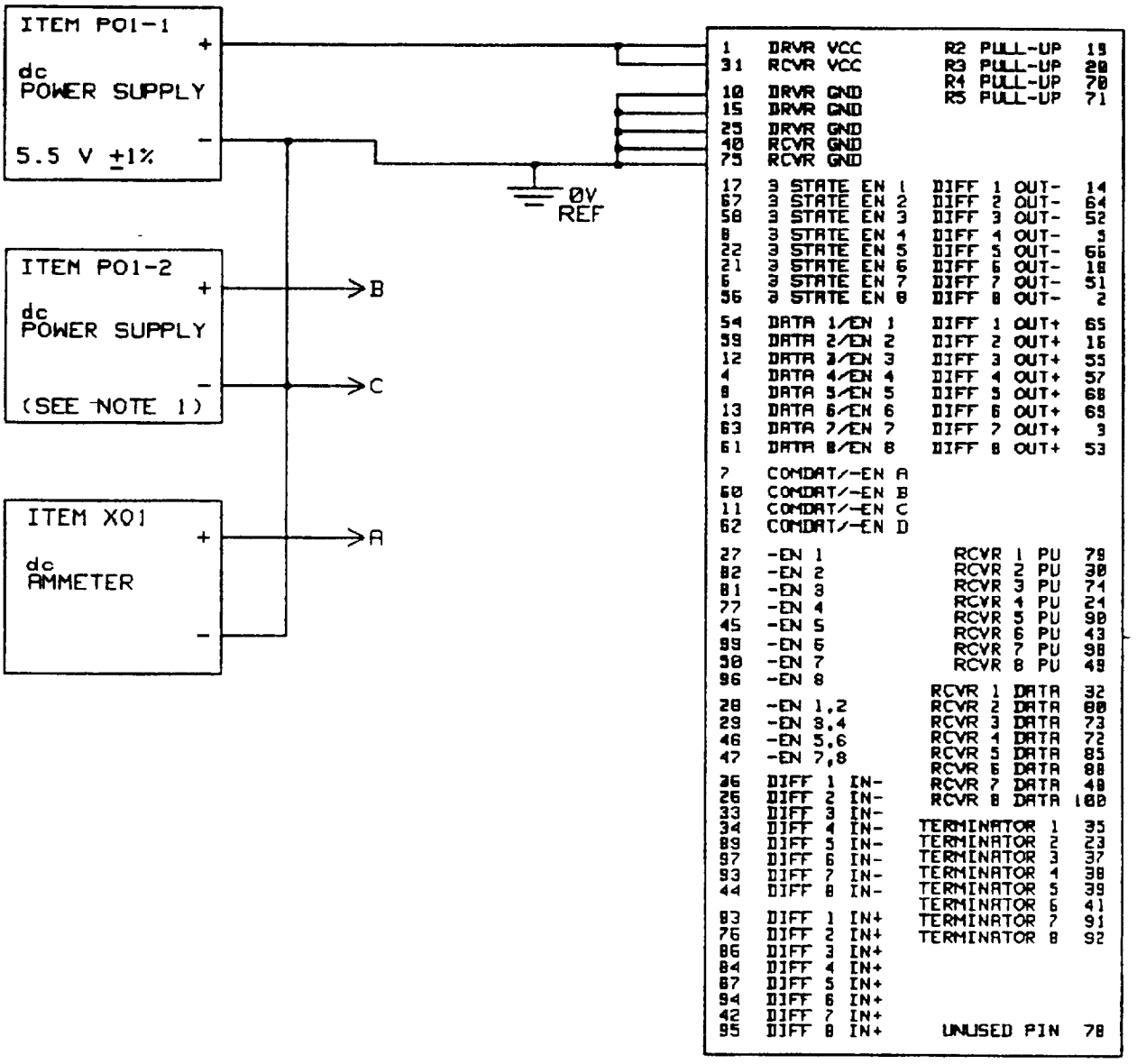


FIGURE 9. Output short circuit current test for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

NOTES:

1. For test conditions:

   D1-D16 - Item PO1-2 = 4.5 V ± 1 percent.
   D17-D24 - Item PO1-2 = 0.5 V ± 1 percent.

2. When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

3. Caution: Do not isolate chassis ground of ammeter XO1 from earth ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE VI. Output leakage current, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E1	DIFF 1 OUT-	14	17	54	-80	80	uA	10
E2	DIFF 2 OUT-	64	67	59				
E3	DIFF 3 OUT-	52	58	12				
E4	DIFF 4 OUT-	5	8	4				
E5	DIFF 5 OUT-	66	22	9				
E6	DIFF 6 OUT-	18	21	13				
E7	DIFF 7 OUT-	51	6	63				
E8	DIFF 8 OUT-	2	56	61				
E9	DIFF 1 OUT+	65	7,17,54	Open				
E10	DIFF 2 OUT+	16	7,67,59					
E11	DIFF 3 OUT+	55	12,58,60					
E12	DIFF 4 OUT+	57	4,8,60					
E13	DIFF 5 OUT+	68	9,11,22					
E14	DIFF 6 OUT+	69	11,13,21					
E15	DIFF 7 OUT+	3	6,62,63					
E16	DIFF 8 OUT+	53	56,61,62	Open	-80	80	uA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE VII Output leakage current, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
F1	DIFF 1 OUT-	14	17,54	Open	-80	80	uA	10
F2	DIFF 2 OUT-	64	59,67					
F3	DIFF 3 OUT-	52	12,58					
F4	DIFF 4 OUT-	5	4,8					
F5	DIFF 5 OUT-	66	9,22					
F6	DIFF 6 OUT-	18	13,21					
F7	DIFF 7 OUT-	51	6,63					
F8	DIFF 8 OUT-	2	56,61	Open				
F9	DIFF 1 OUT+	65	17	7,54				
F10	DIFF 2 OUT+	16	67	7,59				
F11	DIFF 3 OUT+	55	58	12,60				
F12	DIFF 4 OUT+	57	8	4,60				
F13	DIFF 5 OUT+	68	22	9,11				
F14	DIFF 6 OUT+	69	21	11,13				
F15	DIFF 7 OUT+	3	6	62,63	-80	80	uA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE VII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
F16	DIFF 8 OUT+	53	56	61,62	-80	80	uA	10
F17	RCVR 1 DATA	32	83	36	0	200	uA	10
F18	RCVR 2 DATA	80	76	26				
F19	RCVR 3 DATA	73	86	33				
F20	RCVR 4 DATA	72	84	34				
F21	RCVR 5 DATA	85	87	89				
F22	RCVR 6 DATA	88	94	97				
F23	RCVR 7 DATA	48	42	93				
F24	RCVR 8 DATA	100	95	44	0	200	uA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

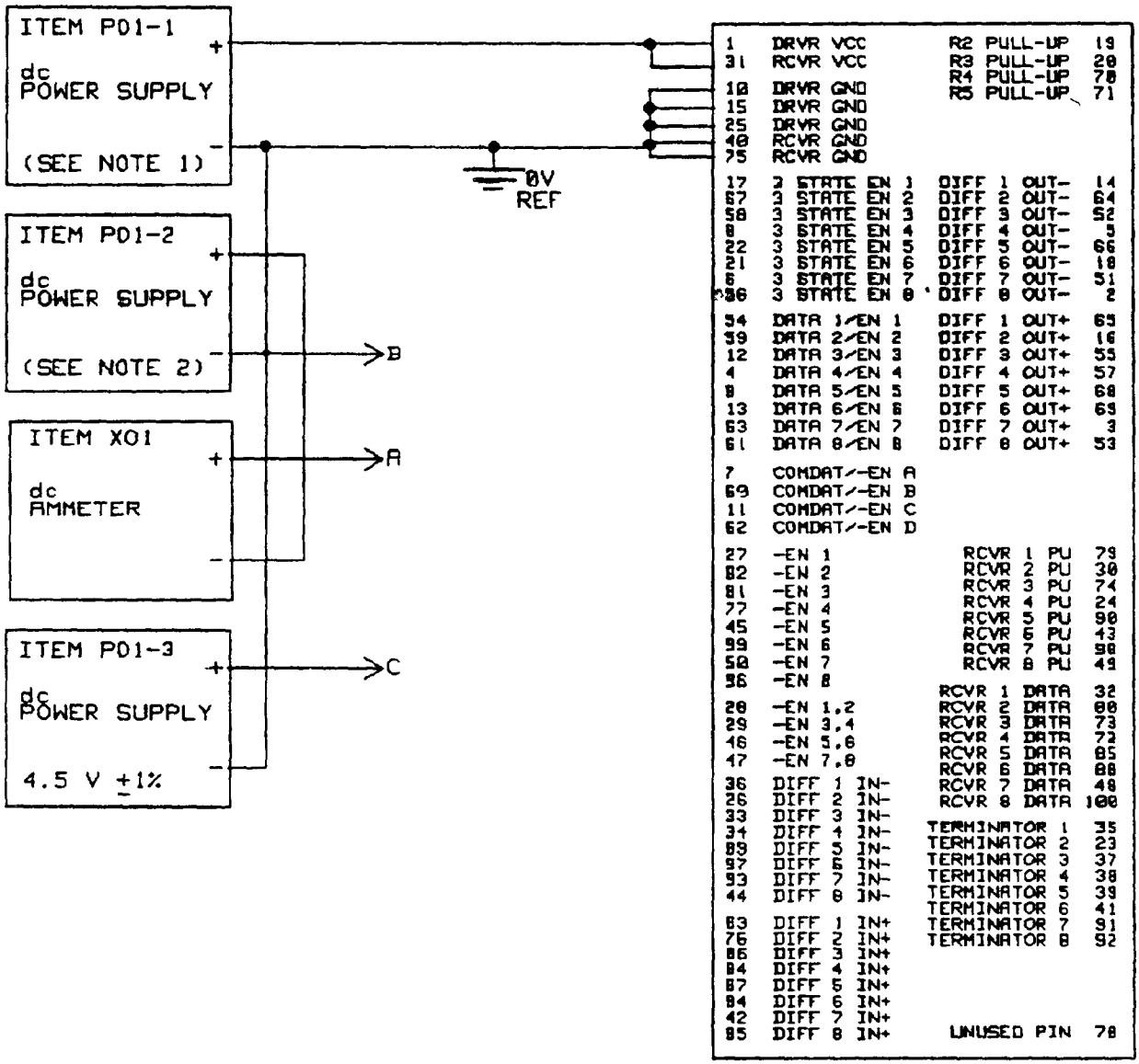


FIGURE 10. Output leakage current test for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

NOTES:

1. For test conditions:

   E1-E16 and F1-F16 - Item PO1-1 = 5.5 V ± 1 percent.
   F17-F24 - Item PO1-1 = 4.5 V ± 1 percent.

2. For test conditions:

   E1-E16 - Item PO1-2 = 0.4 V ± 1 percent.
   F1-F16 - Item PO1-2 = 2.4 V ± 1 percent.
   F17-F24 - Item PO1-2 = 12.0 V ± 1 percent.

3. Caution: Do not isolate the chassis of ammeter XO1 from earth ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test.

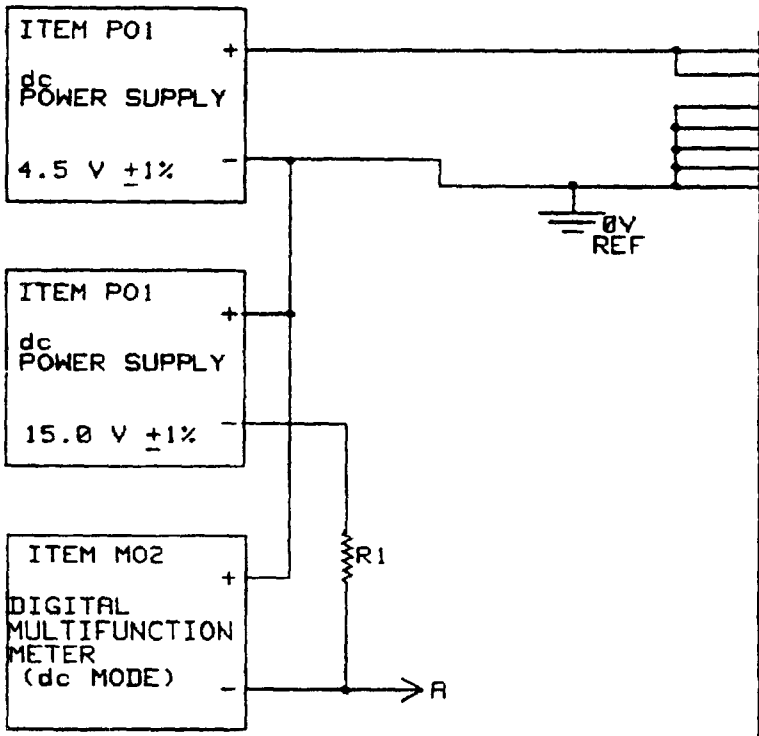
Test cond	Signal name	A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
G1	3 STATE EN 1	17	-0.4	-1.5	V	11
G2	3 STATE EN 2	67				
G3	3 STATE EN 3	58				
G4	3 STATE EN 4	8				
G5	3 STATE EN 5	22				
G6	3 STATE EN 6	21				
G7	3 STATE EN 7	6				
G8	3 STATE EN 8	56				
G9	DATA 1/EN 1	54				
G10	DATA 2/EN 2	59				
G11	DATA 3/EN 3	12				
G12	DATA 4/EN 4	4				
G13	DATA 5/EN 5	9				
G14	DATA 6/EN 6	13				
G15	DATA 7/EN 7	63				
G16	DATA 8/EN 8	61				
G17	COMDAT/-EN A	7				
G18	COMDAT/-EN B	60				
G19	COMDAT/-EN C	11				
G20	COMDAT/-EN D	62	-0.4	-1.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test - Continued.

Test cond	Signal name	A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
G21	-EN 1	27	-0.4	-1.5	V	11
G22	-EN 2	82				
G23	-EN 3	81				
G24	-EN 4	77				
G25	-EN 5	45				
G26	-EN 6	99				
G27	-EN 7	50				
G28	-EN 8	96				
G29	-EN 1,2	28				
G30	-EN 3,4	29				
G31	-EN 5,6	46				
G32	-EN 7,8	47	-0.4	-1.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



1	DRVR VCC	R2 PULL-UP	19
31	RCVR VCC	R3 PULL-UP	20
		R4 PULL-UP	70
10	DRVR GND	R5 PULL-UP	71
15	DRVR GND		
25	DRVR GND		
40	RCVR GND		
75	RCVR GND		
17	3 STATE EN 1	DIFF 1 OUT-	14
67	3 STATE EN 2	DIFF 2 OUT-	64
50	3 STATE EN 3	DIFF 3 OUT-	52
8	3 STATE EN 4	DIFF 4 OUT-	5
22	3 STATE EN 5	DIFF 5 OUT-	66
21	3 STATE EN 6	DIFF 6 OUT-	10
6	3 STATE EN 7	DIFF 7 OUT-	51
56	3 STATE EN 8	DIFF 8 OUT-	2
54	DATA 1/EN 1	DIFF 1 OUT+	65
59	DATA 2/EN 2	DIFF 2 OUT+	16
12	DATA 3/EN 3	DIFF 3 OUT+	55
4	DATA 4/EN 4	DIFF 4 OUT+	57
9	DATA 5/EN 5	DIFF 5 OUT+	68
13	DATA 6/EN 6	DIFF 6 OUT+	69
63	DATA 7/EN 7	DIFF 7 OUT+	3
61	DATA 8/EN 8	DIFF 8 OUT+	53
7	COMDAT/-EN A		
60	COMDAT/-EN B		
11	COMDAT/-EN C		
62	COMDAT/-EN D		
27	-EN 1	RCVR 1 PU	78
82	-EN 2	RCVR 2 PU	30
81	-EN 3	RCVR 3 PU	74
77	-EN 4	RCVR 4 PU	24
45	-EN 5	RCVR 5 PU	90
99	-EN 6	RCVR 6 PU	43
58	-EN 7	RCVR 7 PU	98
96	-EN 8	RCVR 8 PU	49
		RCVR 1 DATA	32
28	-EN 1,2	RCVR 2 DATA	80
29	-EN 3,4	RCVR 3 DATA	73
46	-EN 5,6	RCVR 4 DATA	72
47	-EN 7,8	RCVR 5 DATA	85
		RCVR 6 DATA	88
36	DIFF 1 IN-	RCVR 7 DATA	48
26	DIFF 2 IN-	RCVR 8 DATA	100
33	DIFF 3 IN-		
34	DIFF 4 IN-	TERMINATOR 1	35
89	DIFF 5 IN-	TERMINATOR 2	23
97	DIFF 6 IN-	TERMINATOR 3	37
93	DIFF 7 IN-	TERMINATOR 4	38
44	DIFF 8 IN-	TERMINATOR 5	39
		TERMINATOR 6	41
83	DIFF 1 IN+	TERMINATOR 7	91
76	DIFF 2 IN+	TERMINATOR 8	92
86	DIFF 3 IN+		
84	DIFF 4 IN+		
87	DIFF 5 IN+		
94	DIFF 6 IN+		
42	DIFF 7 IN+		
95	DIFF 8 IN+	UNUSED PIN	78

NOTES:

1. For test conditions G1-G8: R1=1.13 KOhms ± 1 percent, 1/4 W.
2. For test conditions G9-G32: R1=750 Ohms ± 1 percent, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, low level, test.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	25 + 5°C Initial limits Min	Max	Units	Figs
H1	DIFF 1 OUT-	14	17,67	7,54,59	Open	Open	0.0	0.4	V	12,13
H2	DIFF 2 OUT-	64	17,67	7,54,59						
H3	DIFF 3 OUT-	52	8,58	4,12,60						
H4	DIFF 4 OUT-	5	8,58	4,12,60						
H5	DIFF 5 OUT-	66	21,22	9,11,13						
H6	DIFF 6 OUT-	18	21,22	9,11,13						
H7	DIFF 7 OUT-	51	6,56	61,62,63						
H8	DIFF 8 OUT-	2	6,56	61,62,63						
H9	DIFF 1 OUT+	65	17,54,67	7,59						
H10	DIFF 2 OUT+	16	17,59,67	7,54						
H11	DIFF 3 OUT+	55	8,12,58	4,60						
H12	DIFF 4 OUT+	57	4,8,58	12,60						
H13	DIFF 5 OUT+	68	9,21,22	11,13						
H14	DIFF 6 OUT+	69	13,21,22	11,9						
H15	DIFF 7 OUT+	3	6,56,63	61,62	Open	Open	0.0	0.4	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
H16	DIFF B OUT+	53	6,61,56	62,63	Open	Open	0.0	0.4	V	12,13
H17	RCVR 1 PU	32,79	Open	27,28	76,83	26,36				
H18	RCVR 2 PU	30,80		28,82	76,83	26,36				
H19	RCVR 3 PU	73,74		29,81	84,86	33,34				
H20	RCVR 4 PU	24,72		29,77	84,86	33,34				
H21	RCVR 5 PU	85,90		45,46	87,94	89,97				
H22	RCVR 6 PU	43,88		46,99	87,94	89,97				
H23	RCVR 7 PU	48,98		50,47	42,95	44,93				
H24	RCVR 8 PU	49,100	Open	47,96	42,95	44,93	0.0	0.4	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

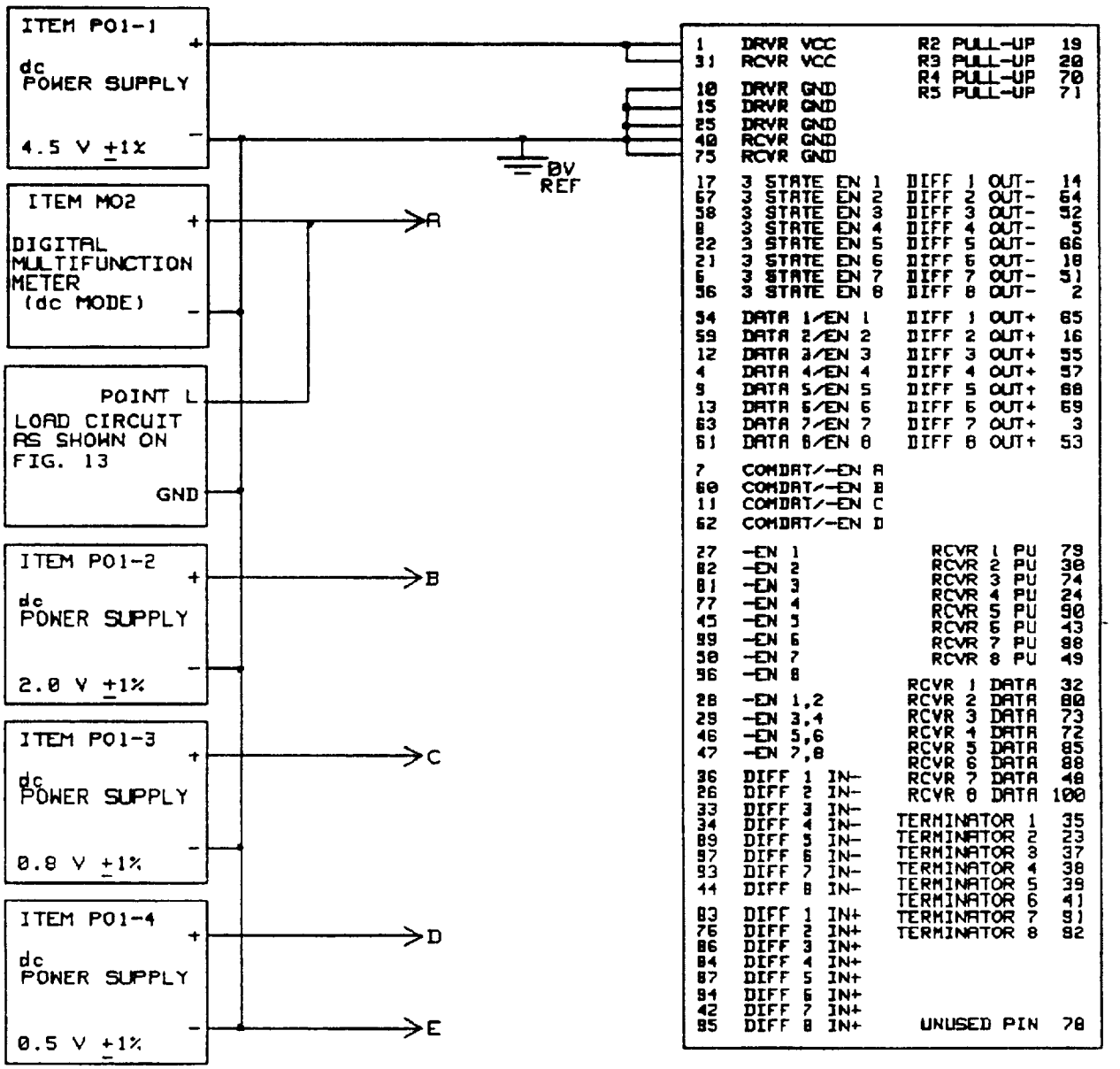
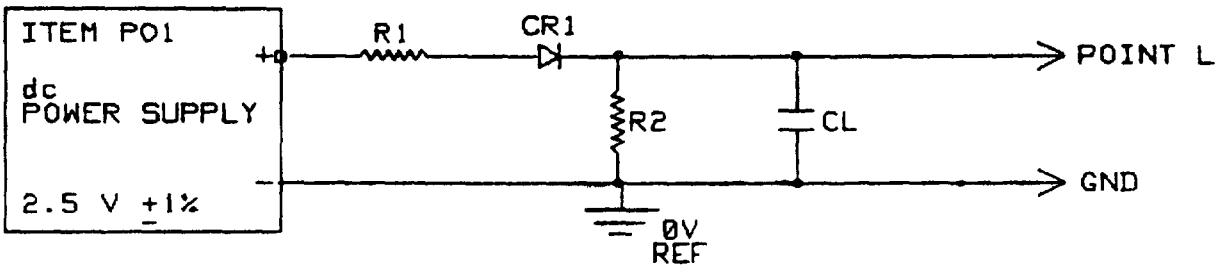


FIGURE 12. Output voltage test circuit for tables IX and X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1=1N3600, 1N4150 or equivalent.

2. For output pins 2,3,5,14,16,18,51-53,55,57,64-66,68,69:

   R1 = Select value for a current flow of 40 mA ± 5 percent with point L held at 0.4 V.
   R2 = 50 Ohms ± 1 percent, 1/8 W. For test conditions H and J.
   R2 = 243 Ohms ± 1 percent, 1/8 W. For all other test conditions.

For output pins (79,32),(30,80),(74,73),(24,72),(90,85),(43,88), (98,48),(49,100):

   R1 = Select value for a current flow of 16 mA ± 5 percent with point L held at 0.4 V.
   R2 = 475 Ohms ± 1 percent, 1/8 W.

This logic load network is applicable only when the RCVR PU output pin is tied to the respective RCVR DATA output pin. Only one load is required per pair.

3. CL = 50 pF (including probe and parasitic capacitance) for crosstalk and enable/disable delay time tests.
   CL = 150 pF (including probe and parasitic capacitance) for sample dynamic and propagation delay time tests.

4. Capacitor CL is not required for dc electrical test conditions H and J.

5. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

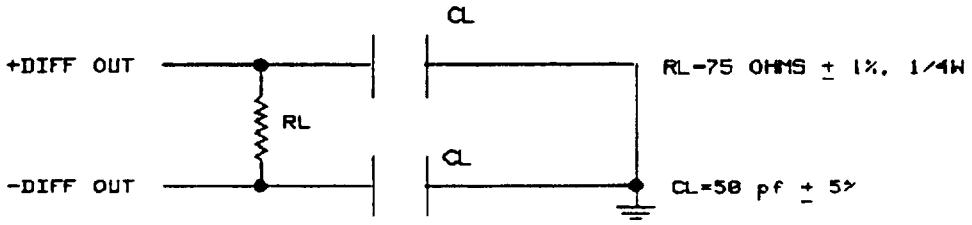
6. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 13. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

7. For test conditions X1-X16 and Y1-Y16, connect the following load to output pins (14,65), (64,16), (52,55), (5,57), (66,68), (18,69), (51,3), (2,53).



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
J1	DIFF 1 OUT-	14	17,54,67	7,59	Open	Open	2.0	4.5	V	12,13
J2	DIFF 2 OUT-	64	17,59,67	7,54						
J3	DIFF 3 OUT-	52	8,12,58	4,60						
J4	DIFF 4 OUT-	5	4,8,58	12,60						
J5	DIFF 5 OUT-	66	9,21,22	11,13						
J6	DIFF 6 OUT-	18	13,21,22	11,9						
J7	DIFF 7 OUT-	51	6,56,63	61,62						
J8	DIFF 8 OUT-	2	6,56,61	62,63						
J9	DIFF 1 OUT+	65	17,67	7,54,59						
J10	DIFF 2 OUT+	16	17,67	7,54,59						
J11	DIFF 3 OUT+	55	8,58	4,12,60						
J12	DIFF 4 OUT+	57	8,58	4,12,60						
J13	DIFF 5 OUT+	68	21,22	9,11,13						
J14	DIFF 6 OUT+	69	21,22	9,11,13						
J15	DIFF 7 OUT+	3	6,56	61,62,63	Open	Open	2.0	4.5	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
J16	DIFF B OUT+	53	6,56	61,62,63	Open	Open	2.0	4.5	V	12,13
J17	RCVR 1 PU	32,79	Open	27,28	26,36	76,83	2.4	4.5	V	12,13
J18	RCVR 2 PU	30,80		28,82	26,36	76,83				
J19	RCVR 3 PU	73,74		29,81	33,34	84,86				
J20	RCVR 4 PU	24,72		29,77	33,34	84,86				
J21	RCVR 5 PU	85,90		45,46	89,97	87,94				
J22	RCVR 6 PU	43,88		46,99	89,97	87,94				
J23	RCVR 7 PU	48,98		50,47	44,93	42,95				
J24	RCVR 8 PU	49,100	Open	47,96	44,93	42,95	2.4	4.5	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K1	3 STATE EN 1	17	Open	Open	-0.7	-1.6	mA	14
K2	3 STATE EN 2	67						
K3	3 STATE EN 3	58						
K4	3 STATE EN 4	8						
K5	3 STATE EN 5	22						
K6	3 STATE EN 6	21						
K7	3 STATE EN 7	6						
K8	3 STATE EN 8	56	Open		-0.7	-1.6	mA	14
K9	DATA 1/EN 1	54	7,4,12,59,60		-30.0	-400.0	uA	14
K10	DATA 2/EN 2	59	7,4,12,54,60					
K11	DATA 3/EN 3	12	7,4,54,59,60					
K12	DATA 4/EN 4	4	7,12,54,59,60					
K13	DATA 5/EN 5	9	11,13,61,62,63					
K14	DATA 6/EN 6	13	9,11,61,62,63					
K15	DATA 7/EN 7	63	9,11,13,61,62	Open	-30.0	-400.0	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K16	DATA 8/EN 8	61	9,11,13,62,63	Open	-30.0	-400.0	uA	14
K17	COMDAT/-EN A	7	4,12,54,59,60					
K18	COMDAT/-EN B	60	4,7,12,54,59					
K19	COMDAT/-EN C	11	9,13,61,62,63					
K20	COMDAT/-EN D	62	9,11,13,61,63					
K21	-EN 1	27	28,29,77,81,82					
K22	-EN 2	82	27,28,29,77,81					
K23	-EN 3	81	27,28,29,77,82					
K24	-EN 4	77	27,28,29,81,82					
K25	-EN 5	45	46,47,50,96,99					
K26	-EN 6	99	45,46,47,50,96					
K27	-EN 7	50	45,46,47,96,99					
K28	-EN 8	96	45,46,47,50,99	Open	-30.0	-400.0	uA	14
K29	-EN 1,2	28	27,29,77,81,82	Open	-60.0	-800.0	uA	14
K30	-EN 3,4	29	27,28,77,81,82	Open	-60.0	-800.0	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K31	-EN 5,6	46	45,47,50,96,99	Open	-60.0	-800.0	uA	14
K32	-EN 7,8	47	45,46,50,96,99		-60.0	-800.0	uA	14
K33	DIFF 1 IN-	36	83		-0.2	-0.9	mA	14
K34	DIFF 2 IN-	26	76					
K35	DIFF 3 IN-	33	86					
K36	DIFF 4 IN-	34	84					
K37	DIFF 5 IN-	89	87					
K38	DIFF 6 IN-	97	94					
K39	DIFF 7 IN-	93	42					
K40	DIFF 8 IN-	44	95					
K41	DIFF 1 IN+	83	36					
K42	DIFF 2 IN+	76	26					
K43	DIFF 3 IN+	86	33					
K44	DIFF 4 IN+	84	34					
K45	DIFF 5 IN+	87	89	Open	-0.2	-0.9	mA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K46	DIFF 6 IN+	94	97	Open	-0.2	-0.9	mA	14
K47	DIFF 7 IN+	42	93					
K48	DIFF 8 IN+	95	44		-0.2	-0.9	mA	14
K49	R2 PULLUP	19	Open		-10.17	-11.47	mA	14
K50	R3 PULLUP	20						
K51	R4 PULLUP	70						
K52	R5 PULLUP	71	Open	Open	-10.17	-11.47	mA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
L1	3 STATE EN 1	17	Open	Remaining input pins	0	40	uA	14
L2	3 STATE EN 2	67						
L3	3 STATE EN 3	58						
L4	3 STATE EN 4	8						
L5	3 STATE EN 5	22						
L6	3 STATE EN 6	21						
L7	3 STATE EN 7	6						
L8	3 STATE EN 8	56			0	40	uA	14
L9	DATA 1/EN 1	54			0	20	uA	14
L10	DATA 2/EN 2	59						
L11	DATA 3/EN 3	12						
L12	DATA 4/EN 4	4						
L13	DATA 5/EN 5	9						
L14	DATA 6/EN 6	13						
L15	DATA 7/EN 7	63	Open	Remaining input pins	0	20	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
L16	DATA 8/EN 8	61	Open	Remaining input pins	0	20	uA	14
L17	COMDAT/-EN A	7						
L18	COMDAT/-EN B	60						
L19	COMDAT/-EN C	11						
L20	COMDAT/-EN D	62						
L21	-EN 1	27						
L22	-EN 2	82						
L23	-EN 3	81						
L24	-EN 4	77						
L25	-EN 5	45						
L26	-EN 6	99						
L27	-EN 7	50						
L28	-EN 8	96			0	20	uA	14
L29	-EN 1,2	28			0	40	uA	14
L30	-EN 3,4	29	Open	Remaining input pins	0	40	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial T limits Min	25 + 5°C Initial T limits Max	Units	Fig
L31	-EN 5,6	46	Open	Remaining input pins	0	40	uA	14
L32	-EN 7,8	47	Open	Remaining input pins	0	40	uA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

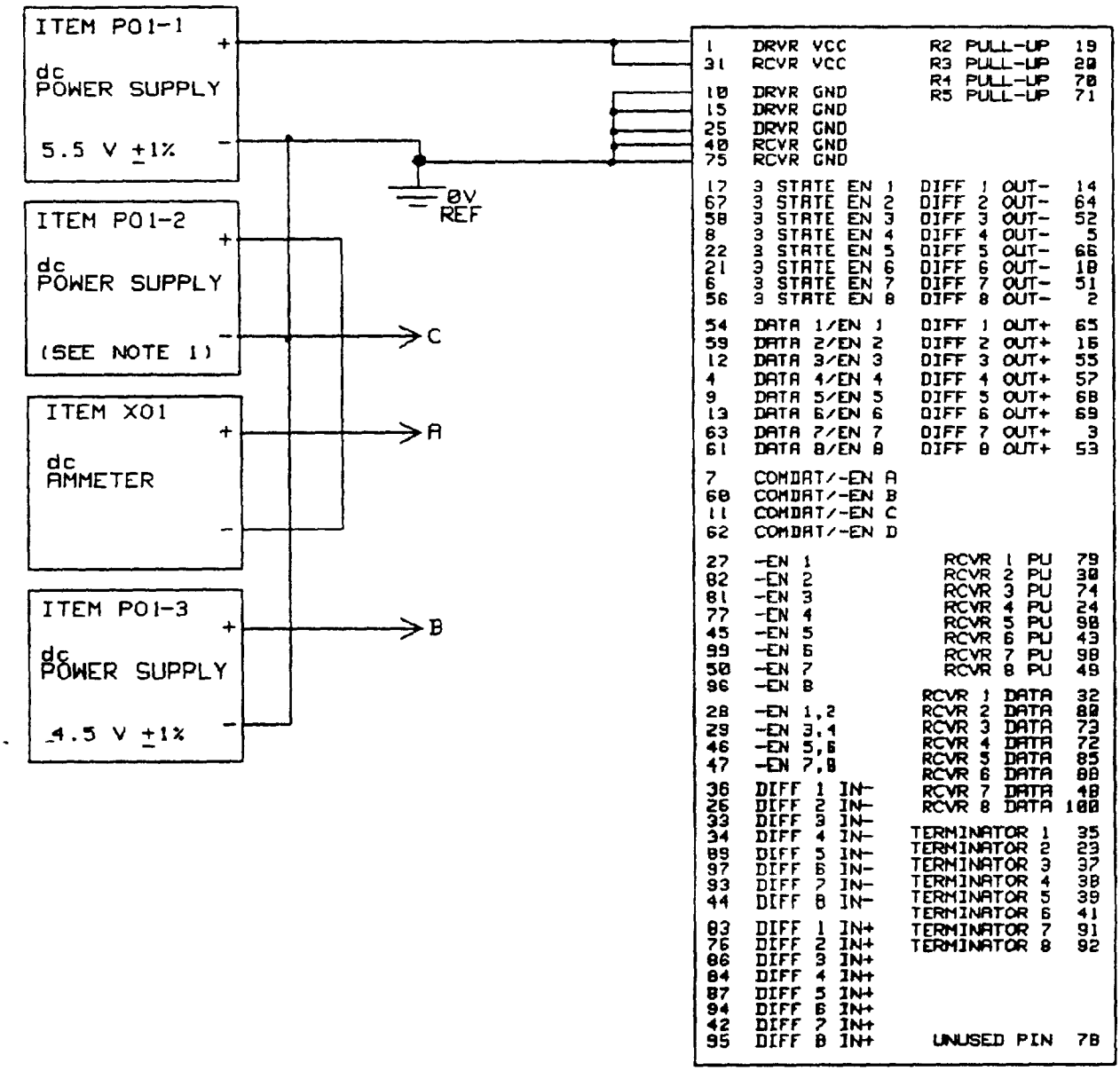


FIGURE 14. Input current test circuit for tables XI and XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

NOTES:

1. For test conditions:

   K1-K48 - Item PO1-2 = 0.4 V ± 1 percent.
   L1-L8 - Item PO1-2 = 2.4 V ± 1 percent.
   L9-L32 - Item PO1-2 = 2.7 V ± 1 percent.
   K49-K52 - Item PO1-2 = 0.0 ± 20 mV.

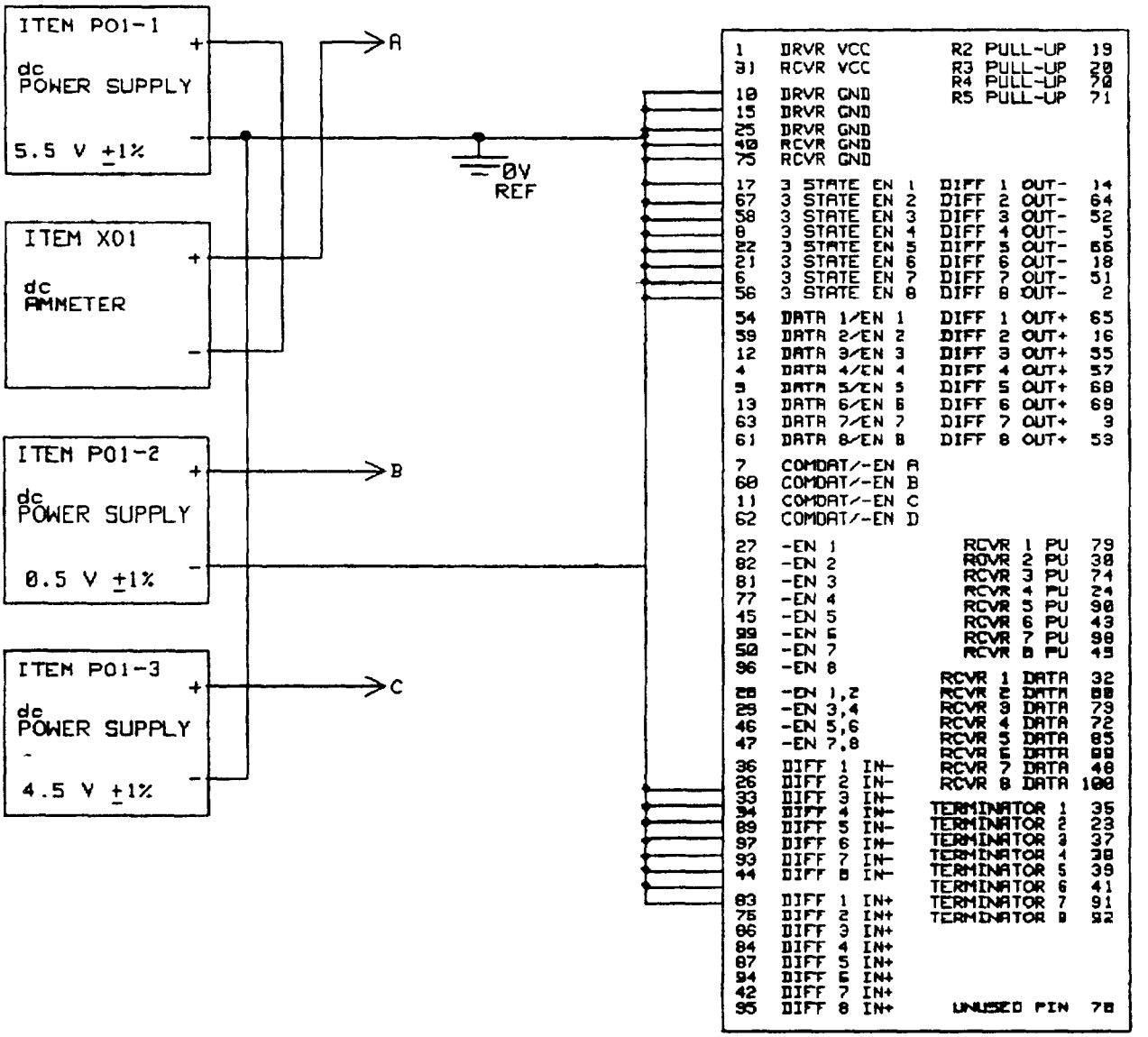
2. Caution: Do not isolate the chassis of ammeter XO1 from earth ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIII. Power supply current (receiver and driver) test.

Test cond	Parameter	Connect			25 + 5°C Initial limits		Units	Fig
		A to pin	B to pin	C to pin	Min	Max		
M1	Icc	31	83,76,86,84, 87,94,42,95	27,82,81,77, 45,99,50,96, 28,29,46,47	21.1	211.0	mA	15
M2	Icc	1	Open	54,59,12,4, 9,13,63,61, 7,60,11,62	28.2	282 0	mA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTE:

1. Caution: Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 15. Power supply current test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N1	DIFF 1 OUT-	14	54	17	7	5	52	nS	16, 17
N2	DIFF 1 OUT-	14	7	17	54	5	52	nS	16, 17
N3	DIFF 2 OUT-	64	59	67	7	5	82	nS	16, 17
N4	DIFF 3 OUT-	52	12	58	60	5	52	nS	16, 17
N5	DIFF 3 OUT-	52	60	58	12	5	52	nS	16, 17
N6	DIFF 4 OUT-	5	4	8	60	5	82	nS	16, 17
N7	DIFF 5 OUT-	66	9	22	11	5	52	nS	16, 17
N8	DIFF 5 OUT-	66	11	22	9	5	52	nS	16, 17
N9	DIFF 6 OUT-	18	13	21	11	5	82	nS	16, 17
N10	DIFF 7 OUT-	51	63	6	62	5	52	nS	16, 17
N11	DIFF 7 OUT-	51	62	6	63	5	52	nS	16, 17
N12	DIFF 8 OUT-	2	61	56	62	5	82	nS	16, 17
N13	DIFF 1 OUT+	65	54	17	7	5	52	nS	16, 17
N14	DIFF 1 OUT+	65	7	17	54	5	52	nS	16, 17
N15	DIFF 2 OUT+	16	59	67	7	5	82	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N16	DIFF 3 OUT+	55	12	58	60	5	52	nS	16, 17
N17	DIFF 3 OUT+	55	60	58	12	5	52	nS	16, 17
N18	DIFF 4 OUT+	57	4	8	60	5	82	nS	16, 17
N19	DIFF 5 OUT+	68	9	22	11	5	52	nS	16, 17
N20	DIFF 5 OUT+	68	11	22	9	5	52	nS	16, 17
N21	DIFF 6 OUT+	69	13	21	11	5	82	nS	16, 17
N22	DIFF 7 OUT+	3	63	6	62	5	52	nS	16, 17
N23	DIFF 7 OUT+	3	62	6	63	5	52	nS	16, 17
N24	DIFF 8 OUT+	53	61	56	62	5	82	nS	16, 17
N25	RCVR 1 PU	32,79	83	Open	27,28,36	5	81	nS	16, 17
N26	RCVR 1 PU	32,79	36	Open	27,28,83	5	81	nS	16, 17
N27	RCVR 1 PU	32,79	27	83	28,36	5	58	nS	16, 17
N28	RCVR 2 PU	30,80	26	Open	28,76,82	5	81	nS	16, 17
N29	RCVR 2 PU	30,80	76	Open	26,28,82	5	81	nS	16, 17
N30	RCVR 2 PU	30,80	82	76	26,28	5	58	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N31	RCVR 3 PU	73,74	86	Open	29,33,81	5	81	nS	16,17
N32	RCVR 3 PU	73,74	33	Open	29,81,86	5	81	nS	16,17
N33	RCVR 3 PU	73,74	81	86	29,33	5	58	nS	16,17
N34	RCVR 4 PU	24,72	84	Open	29,34,77	5	81	nS	16,17
N35	RCVR 4 PU	24,72	34	Open	29,77,84	5	81	nS	16,17
N36	RCVR 4 PU	24,72	77	84	29,34	5	58	nS	16,17
N37	RCVR 5 PU	85,90	87	Open	45,46,89	5	81	nS	16,17
N38	RCVR 5 PU	85,90	89	Open	45,46,87	5	81	nS	16,17
N39	RCVR 5 PU	85,90	45	87	46,89	5	58	nS	16,17
N40	RCVR 6 PU	43,88	94	Open	46,97,99	5	81	nS	16,17
N41	RCVR 6 PU	43,88	97	Open	46,94,99	5	81	nS	16,17
N42	RCVR 6 PU	43,88	99	94	46,97	5	58	nS	16,17
N43	RCVR 7 PU	48,98	42	Open	47,50,93	5	81	nS	16,17
N44	RCVR 7 PU	48,98	93	Open	42,47,50	5	81	nS	16,17
N45	RCVR 7 PU	48,98	50	42	47,93	5	58	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N46	RCVR 8 PU	49,100	95	Open	44,47,96	5	81	nS	16, 17
N47	RCVR 8 PU	49,100	44	Open	47,95,96	5	81	nS	16, 17
N48	RCVR 8 PU	49,100	96	95	44,47	5	58	nS	16, 17
N49	DIFF 2 OUT-	64	7	67	59	4	73	nS	16, 17
N50	DIFF 4 OUT-	5	60	8	4				
N51	DIFF 6 OUT-	18	11	21	13				
N52	DIFF 8 OUT-	2	62	56	61	4	73	nS	16, 17
N53	DIFF 2 OUT+	16	7	67	59	4	82	nS	16, 17
N54	DIFF 4 OUT+	57	60	8	4				
N55	DIFF 6 OUT+	69	11	21	13				
N56	DIFF 8 OUT+	53	62	56	61	4	82	nS	16, 17
N57	RCVR 1 PU	32,79	28	83	27,36	2	58	nS	16, 17
N58	RCVR 2 PU	30,80	28	76	26,82				
N59	RCVR 3 PU	73,74	29	86	33,81				
N60	RCVR 4 PU	24,72	29	84	34,77	2	58	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial T limits Min	25 ± 5°C Initial T limits Max	Units	Figs
N61	RCVR 5 PU	85,90	46	87	45,89	2	58	nS	16, 17
N62	RCVR 6 PU	43,88	46	94	99,97				
N63	RCVR 7 PU	48,98	47	42	50,93				
N64	RCVR 8 PU	49,100	47	95	44,96	2	58	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPLH) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P1	DIFF 1 OUT-	14	54	17	7	5	55	nS	16, 17
P2	DIFF 1 OUT-	14	7	17	54	5	55	nS	16, 17
P3	DIFF 2 OUT-	64	59	67	7	5	85	nS	16, 17
P4	DIFF 3 OUT-	52	12	58	60	5	55	nS	16, 17
P5	DIFF 3 OUT-	52	60	58	12	5	55	nS	16, 17
P6	DIFF 4 OUT-	5	4	8	60	5	85	nS	16, 17
P7	DIFF 5 OUT-	66	9	22	11	5	55	nS	16, 17
P8	DIFF 5 OUT-	66	11	22	9	5	55	nS	16, 17
P9	DIFF 6 OUT-	18	13	21	11	5	85	nS	16, 17
P10	DIFF 7 OUT-	51	63	6	62	5	55	nS	16, 17
P11	DIFF 7 OUT-	51	62	6	63	5	55	nS	16, 17
P12	DIFF 8 OUT-	2	61	56	62	5	85	nS	16, 17
P13	DIFF 1 OUT+	65	54	17	7	5	55	nS	16, 17
P14	DIFF 1 OUT+	65	7	17	54	5	55	nS	16, 17
P15	DIFF 2 OUT+	16	59	67	7	5	85	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
P16	DIFF 3 OUT+	55	12	58	60	5	55	nS	16,17
P17	DIFF 3 OUT+	55	60	58	12	5	55	nS	16,17
P18	DIFF 4 OUT+	57	4	8	60	5	85	nS	16,17
P19	DIFF 5 OUT+	68	9	22	11	5	55	nS	16,17
P20	DIFF 5 OUT+	68	11	22	9	5	55	nS	16,17
P21	DIFF 6 OUT+	69	13	21	11	5	85	nS	16,17
P22	DIFF 7 OUT+	3	63	6	62	5	55	nS	16,17
P23	DIFF 7 OUT+	3	62	6	63	5	55	nS	16,17
P24	DIFF 8 OUT+	53	61	56	62	5	85	nS	16,17
P25	RCVR 1 PU	32,79	83	Open	27,28,36	5	92	nS	16,17
P26	RCVR 1 PU	32,79	36	Open	27,28,83	5	92	nS	16,17
P27	RCVR 1 PU	32,79	27	83	28,36	5	72	nS	16,17
P28	RCVR 2 PU	30,80	26	Open	28,76,82	5	92	nS	16,17
P29	RCVR 2 PU	30,80	76	Open	26,28,82	5	92	nS	16,17
P30	RCVR 2 PU	30,80	82	76	26,28	5	72	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
P31	RCVR 3 PU	73,74	86	Open	29,33,81	5	92	nS	16, 17
P32	RCVR 3 PU	73,74	33	Open	29,81,86	5	92	nS	16, 17
P33	RCVR 3 PU	73,74	81	86	29,33	5	72	nS	16, 17
P34	RCVR 4 PU	24,72	84	Open	29,34,77	5	92	nS	16, 17
P35	RCVR 4 PU	24,72	34	Open	29,77,84	5	92	nS	16, 17
P36	RCVR 4 PU	24,72	77	84	29,34	5	72	nS	16, 17
P37	RCVR 5 PU	85,90	87	Open	45,46,89	5	92	nS	16, 17
P38	RCVR 5 PU	85,90	89	Open	45,46,87	5	92	nS	16, 17
P39	RCVR 5 PU	85,90	45	87	46,89	5	72	nS	16, 17
P40	RCVR 6 PU	43,88	94	Open	46,97,99	5	92	nS	16, 17
P41	RCVR 6 PU	43,88	97	Open	46,94,99	5	92	nS	16, 17
P42	RCVR 6 PU	43,88	99	94	46,97	5	72	nS	16, 17
P43	RCVR 7 PU	48,98	42	Open	47,50,93	5	92	nS	16, 17
P44	RCVR 7 PU	48,98	93	Open	42,47,50	5	92	nS	16, 17
P45	RCVR 7 PU	48,98	50	42	47,93	5	72	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPHL) test - Continued

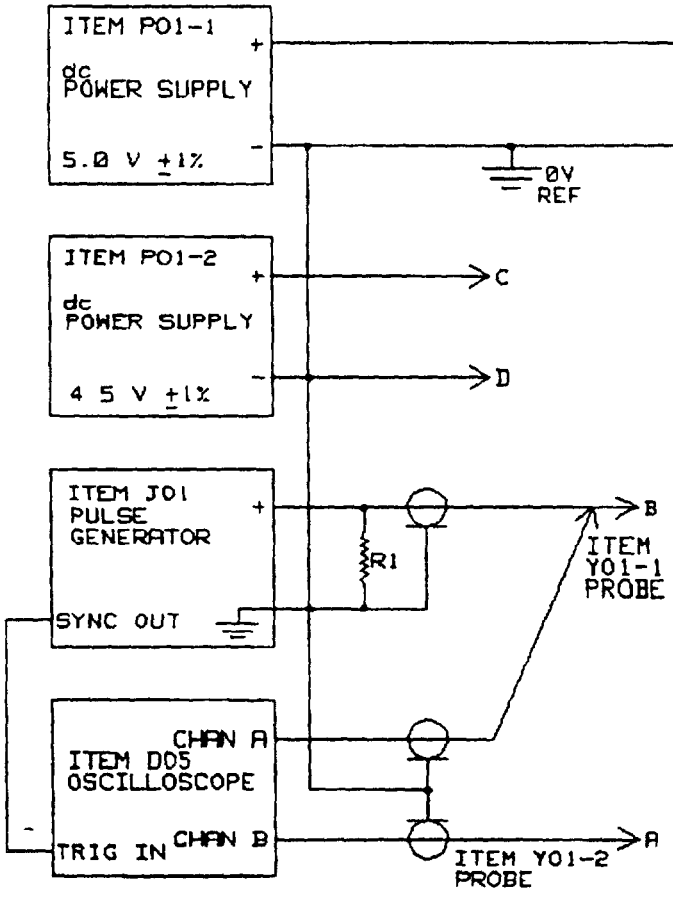
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P46	RCVR 8 PU	49,100	95	Open	44,47,96	5	92	nS	16, 17
P47	RCVR 8 PU	49,100	44	Open	47,95,96	5	92	nS	16, 17
P48	RCVR 8 PU	49,100	96	95	44,47	5	72	nS	16, 17
P49	DIFF 2 OUT-	64	7	67	59	4	85	nS	16, 17
P50	DIFF 4 OUT-	5	60	8	4				
P51	DIFF 6 OUT-	18	11	21	13				
P52	DIFF 8 OUT-	2	62	56	61	4	85	nS	16, 17
P53	DIFF 2 OUT+	16	7	67	59	4	76	nS	16, 17
P54	DIFF 4 OUT+	57	60	8	4				
P55	DIFF 6 OUT+	69	11	21	13				
P56	DIFF 8 OUT+	53	62	56	61	4	76	nS	16, 17
P57	RCVR 1 PU	32,79	28	83	27,36	2	72	nS	16, 17
P58	RCVR 2 PU	30,80	28	76	26,82				
P59	RCVR 3 PU	73,74	29	86	33,81				
P60	RCVR 4 PU	24,72	29	84	34,77	2	72	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial T limits Min	25 ± 5°C Initial T limits Max	Units	Figs
P61	RCVR 5 PU	85,90	46	87	45,89	2	72	nS	16, 17
P62	RCVR 6 PU	43,88	46	94	97,99				
P63	RCVR 7 PU	48,98	47	42	50,93				
P64	RCVR 8 PU	49,100	47	95	44,96	2	72	nS	16, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



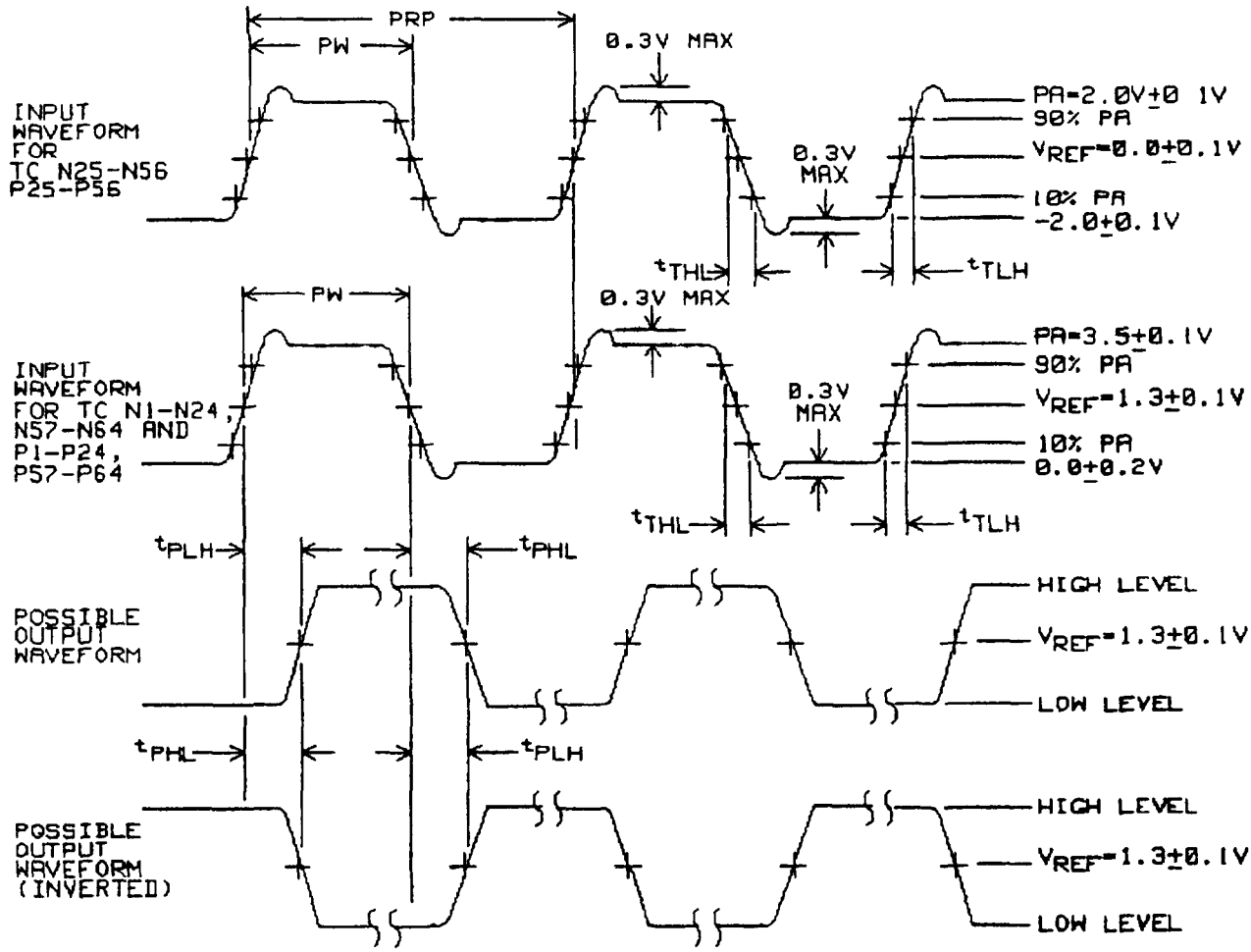
1	DRVR VCC		R2 PULL-UP	19
31	RCVR VCC		R3 PULL-UP	20
			R4 PULL-UP	70
10	DRVR GND		R5 PULL-UP	71
15	DRVR GND			
25	DRVR GND			
40	RCVR GND			
75	RCVR GND			
17	3 STATE EN 1		DIFF 1 OUT-	14
67	3 STATE EN 2		DIFF 2 OUT-	64
58	3 STATE EN 3		DIFF 3 OUT-	52
8	3 STATE EN 4		DIFF 4 OUT-	5
22	3 STATE EN 5		DIFF 5 OUT-	66
21	3 STATE EN 6		DIFF 6 OUT-	18
6	3 STATE EN 7		DIFF 7 OUT-	51
56	3 STATE EN 8		DIFF 8 OUT-	2
54	DATA 1/EN 1		DIFF 1 OUT+	65
59	DATA 2/EN 2		DIFF 2 OUT+	16
12	DATA 3/EN 3		DIFF 3 OUT+	55
4	DATA 4/EN 4		DIFF 4 OUT+	57
9	DATA 5/EN 5		DIFF 5 OUT+	68
13	DATA 6/EN 6		DIFF 6 OUT+	69
63	DATA 7/EN 7		DIFF 7 OUT+	3
61	DATA 8/EN 8		DIFF 8 OUT+	53
7	COMDAT/-EN A			
60	COMDAT/-EN B			
11	COMDAT/-EN C			
62	COMDAT/-EN D			
27	-EN 1		RCVR 1 PU	79
82	-EN 2		RCVR 2 PU	30
81	-EN 3		RCVR 3 PU	74
77	-EN 4		RCVR 4 PU	24
45	-EN 5		RCVR 5 PU	90
99	-EN 6		RCVR 6 PU	43
50	-EN 7		RCVR 7 PU	98
96	-EN 8		RCVR 8 PU	49
			RCVR 1 DATA	32
28	-EN 1,2		RCVR 2 DATA	80
29	-EN 3,4		RCVR 3 DATA	73
46	-EN 5,6		RCVR 4 DATA	72
47	-EN 7,8		RCVR 5 DATA	85
			RCVR 6 DATA	88
36	DIFF 1 IN-		RCVR 7 DATA	48
26	DIFF 2 IN-		RCVR 8 DATA	100
33	DIFF 3 IN-			
34	DIFF 4 IN-		TERMINATOR 1	35
89	DIFF 5 IN-		TERMINATOR 2	23
97	DIFF 6 IN-		TERMINATOR 3	37
93	DIFF 7 IN-		TERMINATOR 4	38
44	DIFF 8 IN-		TERMINATOR 5	39
			TERMINATOR 6	41
83	DIFF 1 IN+		TERMINATOR 7	91
76	DIFF 2 IN+		TERMINATOR 8	92
86	DIFF 3 IN+			
84	DIFF 4 IN+			
87	DIFF 5 IN+			
94	DIFF 6 IN+			
42	DIFF 7 IN+			
95	DIFF 8 IN+		UNUSED PIN	78

NOTES:

1. R1 = 51.1 Ohms ± 5 percent.
2. Load each output as specified on figure 13.

FIGURE 16. Propagation delay time test circuit for tables XIV and XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: PW = 500 ± 50 nS, PRP = 1000 ± 100 nS, tTHL = tTLH ≤ 5 nS.

2. Before taking the measurements, verify that the output pulse high level is ≥ 2.0 V and the low level is ≤ 0.5 V; and the receiver high level is ≥ 2.4 V and the low level is ≤ 0.5 V.

FIGURE 17. Propagation delay time waveforms for tables XIV and XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output enable delay time (tPZH) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Q1	DIFF 1 OUT-	14	17	54	7	2	27	nS	18, 19
Q2	DIFF 2 OUT-	64	67	59	7				
Q3	DIFF 3 OUT-	52	58	12	60				
Q4	DIFF 4 OUT-	5	8	4	60				
Q5	DIFF 5 OUT-	66	22	9	11				
Q6	DIFF 6 OUT-	18	21	13	11				
Q7	DIFF 7 OUT-	51	6	63	62				
Q8	DIFF 8 OUT-	2	56	61	62				
Q9	DIFF 1 OUT+	65	17	Open	7,54				
Q10	DIFF 2 OUT+	16	67		7,59				
Q11	DIFF 3 OUT+	55	58		12,60				
Q12	DIFF 4 OUT+	57	8		4,60				
Q13	DIFF 5 OUT+	68	22		9,11				
Q14	DIFF 6 OUT+	69	21		11,13				
Q15	DIFF 7 OUT+	3	6	Open	62,63	2	27	nS	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output enable time (tPZH) test - Continued.

Test cond	Signal name	Connect				25 ± 5°C Initial limits		Units	Figs
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q16	DIFF B OUT+	53	56	Open	61,62	2	27	nS	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output disable delay time (tPHZ) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
R1	DIFF 1 OUT-	14	17	54	7	2	25	nS	18, 19
R2	DIFF 2 OUT-	64	67	59	7				
R3	DIFF 3 OUT-	52	58	12	60				
R4	DIFF 4 OUT-	5	8	4	60				
R5	DIFF 5 OUT-	66	22	9	11				
R6	DIFF 6 OUT-	18	21	13	11				
R7	DIFF 7 OUT-	51	6	63	62				
R8	DIFF 8 OUT-	2	56	61	62				
R9	DIFF 1 OUT+	65	17	Open	7,54				
R10	DIFF 2 OUT+	16	67		7,59				
R11	DIFF 3 OUT+	55	58		12,60				
R12	DIFF 4 OUT+	57	8		4,60				
R13	DIFF 5 OUT+	68	22		9,11				
R14	DIFF 6 OUT+	69	21		11,13				
R15	DIFF 7 OUT+	3	6	Open	62,63	2	25	nS	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output disable time (tPHZ) test - Continued.

Test cond	Signal name	Connect				25 + 5°C Initial limits		Units	Figs
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R16	DIFF B OUT+	53	56	Open	61,62	2	25	nS	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Output enable delay time (tPZL) test.

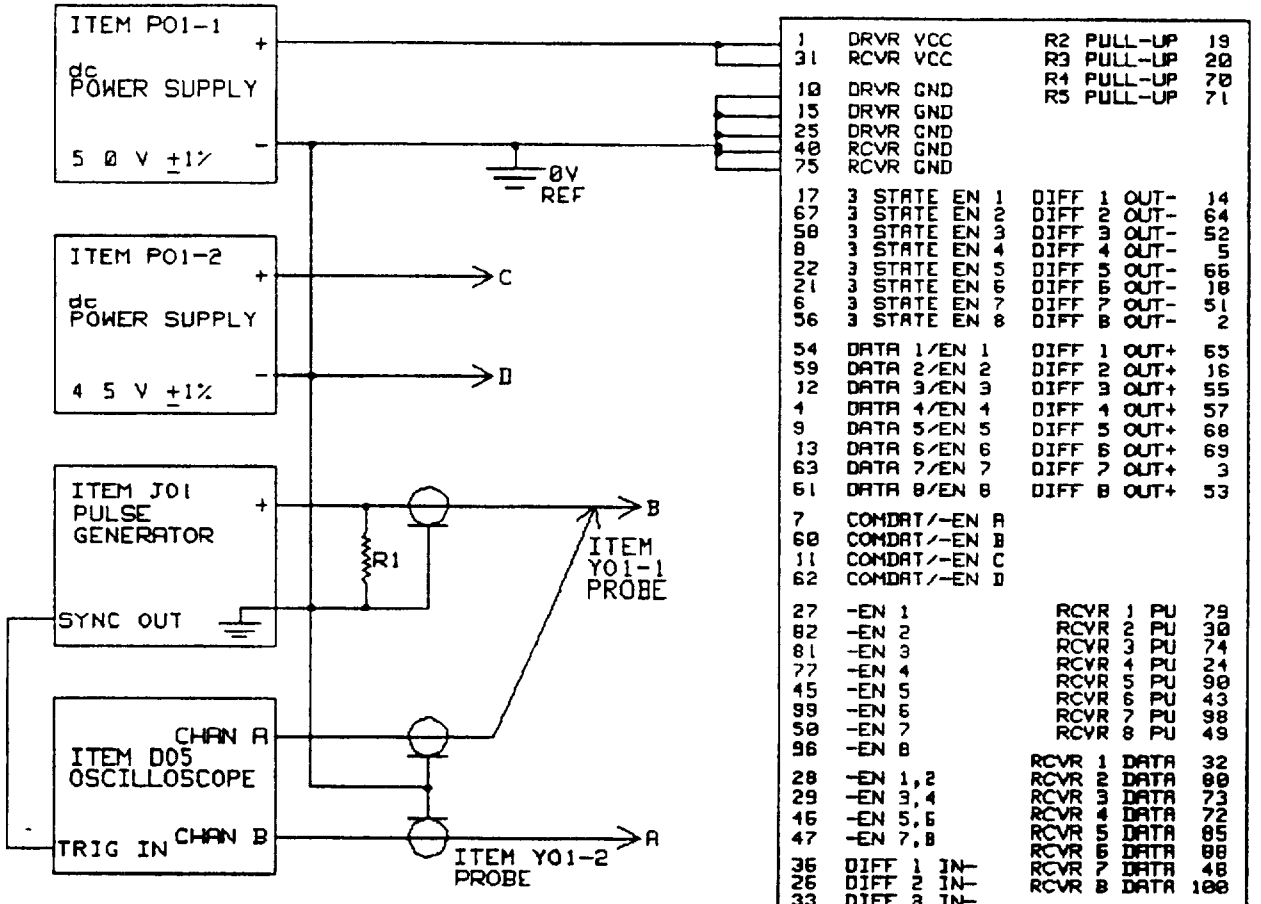
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
S1	DIFF 1 OUT-	14	17	Open	7,54	2	38	nS	18,19
S2	DIFF 2 OUT-	64	67		7,59				
S3	DIFF 3 OUT-	52	58		12,60				
S4	DIFF 4 OUT-	5	8		4,60				
S5	DIFF 5 OUT-	66	22		9,11				
S6	DIFF 6 OUT-	18	21		11,13				
S7	DIFF 7 OUT-	51	6		62,63				
S8	DIFF 8 OUT-	2	56	Open	61,62				
S9	DIFF 1 OUT+	65	17	54	7				
S10	DIFF 2 OUT+	16	67	59	7				
S11	DIFF 3 OUT+	55	58	12	60				
S12	DIFF 4 OUT+	57	8	60	4				
S13	DIFF 5 OUT+	68	22	9	11				
S14	DIFF 6 OUT+	69	21	11	13				
S15	DIFF 7 OUT+	3	6	63	62				
S16	DIFF 8 OUT+	53	56	62	61	2	38	nS	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XIX. Output disable delay time (tPLZ) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
T1	DIFF 1 OUT-	14	17	Open	7,54	2	47	nS	18,19
T2	DIFF 2 OUT-	64	67		7,59				
T3	DIFF 3 OUT-	52	58		12,60				
T4	DIFF 4 OUT-	5	8		4,60				
T5	DIFF 5 OUT-	66	22		9,11				
T6	DIFF 6 OUT-	18	21		11,13				
T7	DIFF 7 OUT-	51	6		62,63				
T8	DIFF 8 OUT-	2	56	Open	61,62				
T9	DIFF 1 OUT+	65	17	54	7				
T10	DIFF 2 OUT+	16	67	59	7				
T11	DIFF 3 OUT+	55	58	12	60				
T12	DIFF 4 OUT+	57	8	60	4				
T13	DIFF 5 OUT+	68	22	9	11				
T14	DIFF 6 OUT+	69	21	11	13				
T15	DIFF 7 OUT+	3	6	63	62				
T16	DIFF 8 OUT+	53	56	62	61	2	47	nS	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

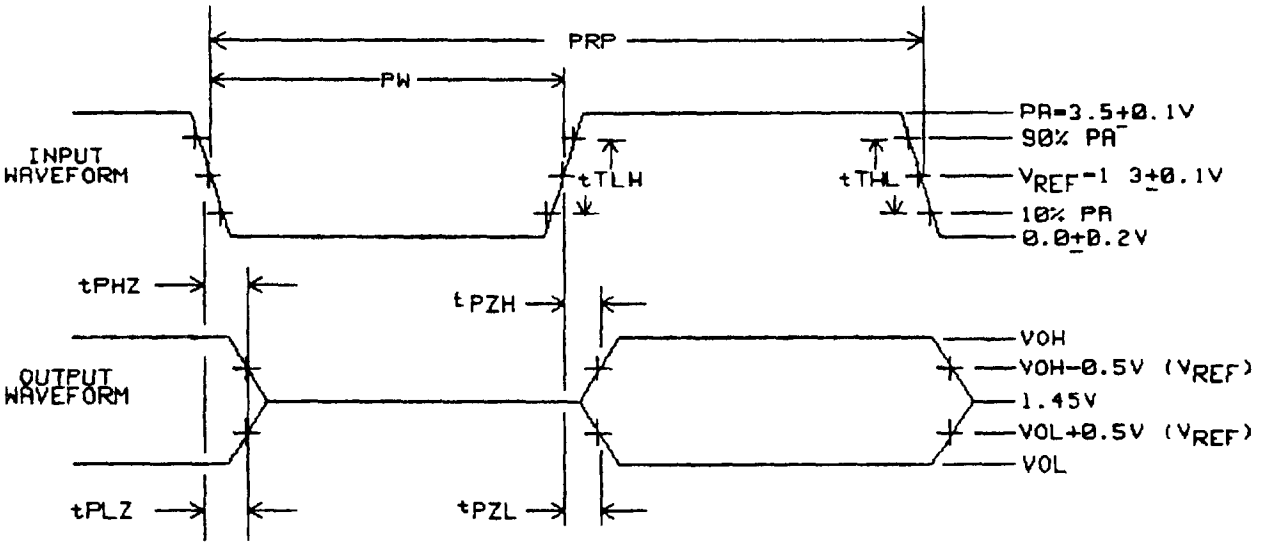


NOTES:

1. R1 = 51.1 Ohms ± 5 percent.
2. Load each output as specified on figure 13.

FIGURE 18. Output enable/disable delay time test circuit for tables XVI, XVII, XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: PW = 500 ± 50 nS, PRP = 1000 ± 100 nS, tTHL = tTLH ≤ 5 nS.

2. Before taking the measurements, verify the output pulse high is ≥ 2.0 V and the low level is ≤ 0.5 V.

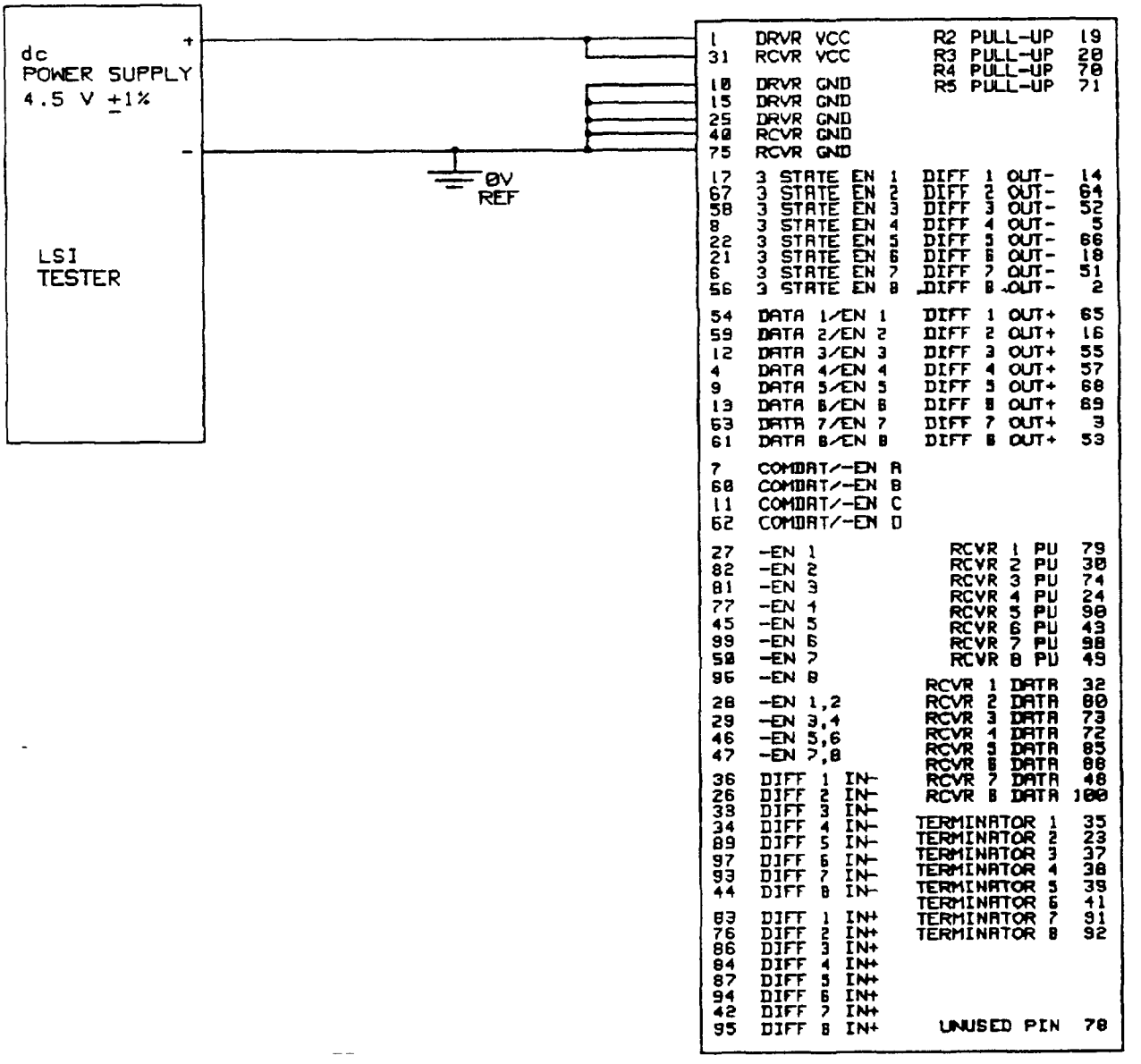
FIGURE 19. Output enable/disable delay time test waveforms for tables XVI, XVII, XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XX. Functional test.

Test cond	Test description	Fig
V	Functional test	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input conditioning voltage shall be 0.4 V ± 1 percent for logic 0, and 2.4 V ± 1 percent for logic 1.

2. Voh = 2.4 V min, Vol = 0.4 V max.

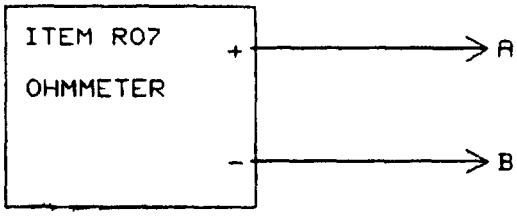
FIGURE 20. Functional test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXI. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
W1	Continuity	10	15	0	1.0	Ohms	21
W2	Continuity	10	75				
W3	Continuity	40	25	0	1.0	Ohms	21
W4	Continuity	10	15	0	0.1	Ohms	21
W5	Continuity	10	75				
W6	Continuity	40	25	0	0.1	Ohms	21

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



1	DRVR VCC	R2 PULL-UP	19
31	RCVR VCC	R3 PULL-UP	20
		R4 PULL-UP	70
10	DRVR GND	R5 PULL-UP	71
15	DRVR GND		
25	DRVR GND		
40	RCVR GND		
75	RCVR GND		
17	3 STATE EN 1	DIFF 1 OUT-	14
67	3 STATE EN 2	DIFF 2 OUT-	64
58	3 STATE EN 3	DIFF 3 OUT-	52
8	3 STATE EN 4	DIFF 4 OUT-	5
22	3 STATE EN 5	DIFF 5 OUT-	66
21	3 STATE EN 6	DIFF 6 OUT-	18
6	3 STATE EN 7	DIFF 7 OUT-	51
56	3 STATE EN 8	DIFF 8 OUT-	2
54	DATA 1/EN 1	DIFF 1 OUT+	65
59	DATA 2/EN 2	DIFF 2 OUT+	16
12	DATA 3/EN 3	DIFF 3 OUT+	55
4	DATA 4/EN 4	DIFF 4 OUT+	57
9	DATA 5/EN 5	DIFF 5 OUT+	68
13	DATA 6/EN 6	DIFF 6 OUT+	69
63	DATA 7/EN 7	DIFF 7 OUT+	3
61	DATA 8/EN 8	DIFF 8 OUT+	53
7	COMDAT/-EN A		
60	COMDAT/-EN B		
11	COMDAT/-EN C		
62	COMDAT/-EN D		
27	-EN 1	RCVR 1 PU	79
82	-EN 2	RCVR 2 PU	30
81	-EN 3	RCVR 3 PU	74
77	-EN 4	RCVR 4 PU	24
45	-EN 5	RCVR 5 PU	90
99	-EN 6	RCVR 6 PU	43
50	-EN 7	RCVR 7 PU	98
96	-EN 8	RCVR 8 PU	49
		RCVR 1 DATA	32
28	-EN 1,2	RCVR 2 DATA	80
29	-EN 3,4	RCVR 3 DATA	79
46	-EN 5,6	RCVR 4 DATA	72
47	-EN 7,8	RCVR 5 DATA	85
		RCVR 6 DATA	88
36	DIFF 1 IN-	RCVR 7 DATA	48
26	DIFF 2 IN-	RCVR 8 DATA	100
33	DIFF 3 IN-		
34	DIFF 4 IN-	TERMINATOR 1	35
89	DIFF 5 IN-	TERMINATOR 2	23
97	DIFF 6 IN-	TERMINATOR 3	37
93	DIFF 7 IN-	TERMINATOR 4	38
44	DIFF 8 IN-	TERMINATOR 5	39
		TERMINATOR 6	41
83	DIFF 1 IN+	TERMINATOR 7	91
76	DIFF 2 IN+	TERMINATOR 8	92
88	DIFF 3 IN+		
84	DIFF 4 IN+		
87	DIFF 5 IN+		
94	DIFF 6 IN+		
42	DIFF 7 IN+		
95	DIFF 8 IN+	UNUSED PIN	70

FIGURE 21. Continuity test circuit for table XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk, low level, test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
X1	DIFF 1 OUT-	65	14	4,9,12,13, 27,45,50,59, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95	7,11,26,28, 29,33,34,36, 44,46,47,54, 60,62,89,93, 97	0	400	mV	22,23
X2	DIFF 2 OUT-	16	64	4,9,12,13, 27,45,50,54, 61,63,77,81, 82,96,99		7,11,26,28, 29,33,34,36, 44,46,47,59, 60,62,89,93, 97				
X3	DIFF 3 OUT-	55	52	4,9,13,27, 45,50,54,59, 61,63,77,81, 82,96,99		7,11,12,26, 28,29,33,34, 36,44,46,47, 60,62,89,93, 97				
X4	DIFF 4 OUT-	57	5	9,12,13,27, 45,50,54,59, 61,63,77,81, 82,96,99		4,7,11,26, 28,29,33,34, 36,44,46,47, 60,62,89,93, 97				
X5	DIFF 5 OUT-	68	66	4,12,13,27, 45,50,54,59, 61,63,77,81, 82,96,98		7,9,11,26, 28,29,33,34, 36,44,46,47, 60,62,89,93, 97				
X6	DIFF 6 OUT-	68	18	4,9,12,27, 45,50,54,59, 61,63,77,81, 82,96,98		7,11,13,26, 28,29,33,34, 36,44,46,47, 60,62,89,93, 97				
X7	DIFF 7 OUT-	3	51	4,9,12,13, 27,45,50,54, 59,61,77,81, 82,96,98		7,11,26,28, 29,33,34,36, 44,46,47,60, 62,63,89,93, 97				
X8	DIFF 8 OUT-	53	2	4,9,12,13, 27,45,50,54, 59,63,77,81, 82,96,98	6,8,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95	7,11,26,28, 29,33,34,36, 44,46,47,60, 62,63,89,93, 97	0	400	mV	22,23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk, low level, test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
X9	RCVR 1 PU	32, 79	Open	4,9,12,13, 45,50,54,59, 61,63,77,81, 82,96,99	17,67,58,22, 21,6,56,94, 42,83,76,95, 86,84,87	7,11,26,27, 28,29,33,34, 36,44,46,47, 60,62,89,93, 97	0	400	mV	22,23
X10	RCVR 2 PU	30, 80		4,9,12,13, 27,45,50,54, 59,61,63,77, 81,96,99		7,11,26,28, 29,33,34,36, 44,46,47,60, 82,62,89,93, 97				
X11	RCVR 3 PU	73, 74		4,9,12,13, 27,45,50,54, 59,61,63,77, 82,96,99		7,11,26,28, 29,33,34,36, 44,46,47,60, 81,62,89,93, 97				
X12	RCVR 4 PU	24, 72		4,9,12,13, 27,45,50,54, 59,61,63,81, 82,96,99		7,11,26,28, 29,33,34,36, 44,46,47,60, 77,62,89,93, 97				
X13	RCVR 5 PU	85, 90		4,9,12,13, 27,50,54,59, 61,63,77,81, 82,96,99		7,11,26,28, 29,33,34,36, 45,44,46,47, 60,62,89,93, 97				
X14	RCVR 6 PU	43, 88		4,9,12,13, 27,45,50,54, 59,61,63,77, 81,82,96		7,11,26,28, 29,33,34,36, 44,46,47,60, 62,89,93,97, 99				
X15	RCVR 7 PU	48, 98		4,9,12,13, 27,45,54,59, 61,63,77,81, 82,96,99		7,11,26,28, 29,33,34,36, 44,46,47,60, 50,62,89,93, 97				
X16	RCVR 8 PU	49, 100	Open	4,9,12,13, 27,45,50,54, 59,61,63,77, 81,82,99	17,67,58,22, 21,6,56,94, 42,83,76,95, 86,84,87	7,11,6,28, 29,33,34,36, 44,46,47,60, 62,89,93,97,	0	400	mV	22,23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Output crosstalk, high level, test.

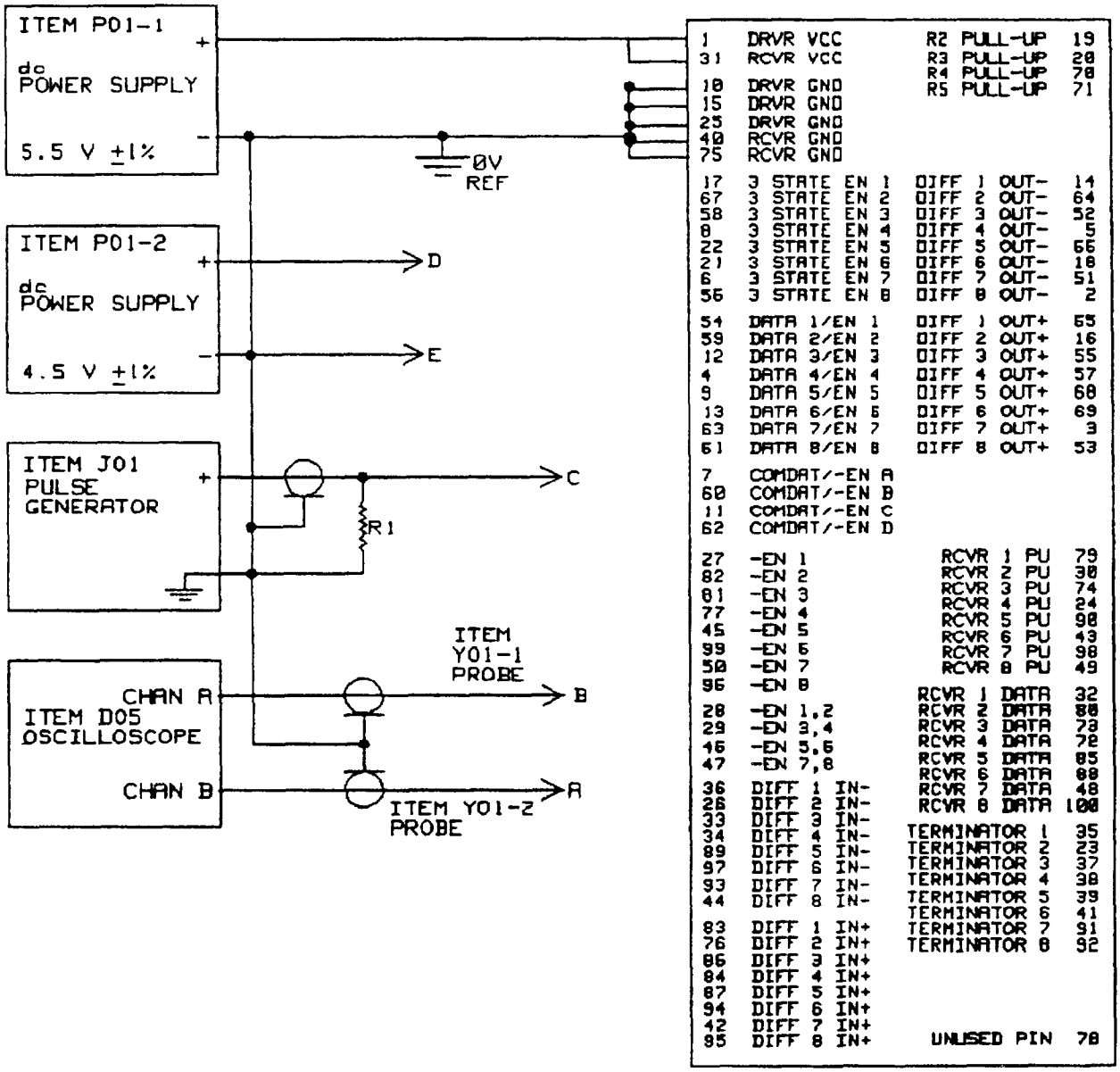
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Y1	DIFF 1 OUT-	65	14	4,9,12,13, 27,45,50,59, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,54,56, 58,67,76,83, 84,86,87,94, 95	7,11,26,28, 29,33,34,36, 44,46,47,60, 62,89,93,97	0	400	mV	22,23
Y2	DIFF 2 OUT-	16	64	6,9,12,13, 27,45,50,54, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,56,58, 59,67,76,83, 84,86,87,94, 95					
Y3	DIFF 3 OUT-	55	52	4,9,13,27, 45,50,54,59, 61,63,77,81, 82,96,99	6,8,12,17, 21,22,42,56, 58,67,76,83, 84,86,87,94, 95					
Y4	DIFF 4 OUT-	57	5	9,12,13,27, 45,50,54,59, 61,63,77,81, 82,96,99	4,6,8,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95					
Y5	DIFF 5 OUT-	68	66	4,12,13,27, 45,50,54,59, 61,63,77,81, 82,96,99	6,8,9,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95					
Y6	DIFF 6 OUT-	69	18	4,9,12,27, 45,50,54,59, 61,63,77,81, 82,96,99	6,8,13,17, 21,22,42,56, 58,67,76,83, 84,86,87,94, 95					
Y7	DIFF 7 OUT-	3	51	4,9,12,13, 27,45,50,54, 59,61,77,81, 82,96,99	6,8,17,21, 22,42,56,58, 63,67,76,83, 84,86,87,94, 95					
Y8	DIFF 8 OUT-	53	2	4,9,12,13, 27,45,50,54, 59,63,77,81, 82,96,99	6,8,17,21, 22,42,56,58, 61,67,76,83, 84,86,87,94, 95	7,11,26,28, 29,33,34,36, 44,46,47,60, 62,89,93,97	0	400	mV	22,23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXIII Output crosstalk, high level, test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Y9	RCVR 1 PU	32, 79	Open	4,9,12,13, 45,50,54,59, 61,63,77,81, 82,98,99	6,8,17,21, 22,27,42,56, 58,67,76,83, 84,86,87,94, 95	7,11,26,28, 29,33,34,36, 44,46,47,60, 62,89,93,97	0	400	mV	22,23
Y10	RCVR 2 PU	30, 80		4,9,12,13, 27,45,50,54, 59,61,63,77, 81,96,99	6,8,17,21, 22,42,56,58, 67,76,82,83, 84,86,87,94, 95					
Y11	RCVR 3 PU	73, 74		4,9,13,27, 45,50,54,59, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,56,58, 67,76,81,83, 84,86,87,94, 95					
Y12	RCVR 4 PU	24, 72		4,9,12,13, 27,45,50,54, 59,61,63,81, 82,96,99	6,8,17,21, 22,42,56,58, 67,76,77,83, 84,86,87,94, 95					
Y13	RCVR 5 PU	85, 90		4,9,12,13, 27,50,54,59, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,45,56, 58,67,76,83, 84,86,87,94, 95					
Y14	RCVR 6 PU	43, 88		4,9,12,13, 27,45,50,54, 59,61,83,77, 81,82,96	6,8,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95, 99					
Y15	RCVR 7 PU	48, 98		4,9,12,13, 27,45,54,59, 61,63,77,81, 82,96,99	6,8,17,21, 22,42,50,56, 58,67,76,83, 84,86,87,94, 95					
Y16	RCVR 8 PU	49, 100	Open	4,9,12,13, 27,45,50,54, 59,61,83,77, 81,82,99	6,8,17,21, 22,42,56,58, 67,76,83,84, 86,87,94,95, 98	7,11,26,28, 29,33,34,36, 44,46,47,60, 62,89,93,97	0	400	mV	22,23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

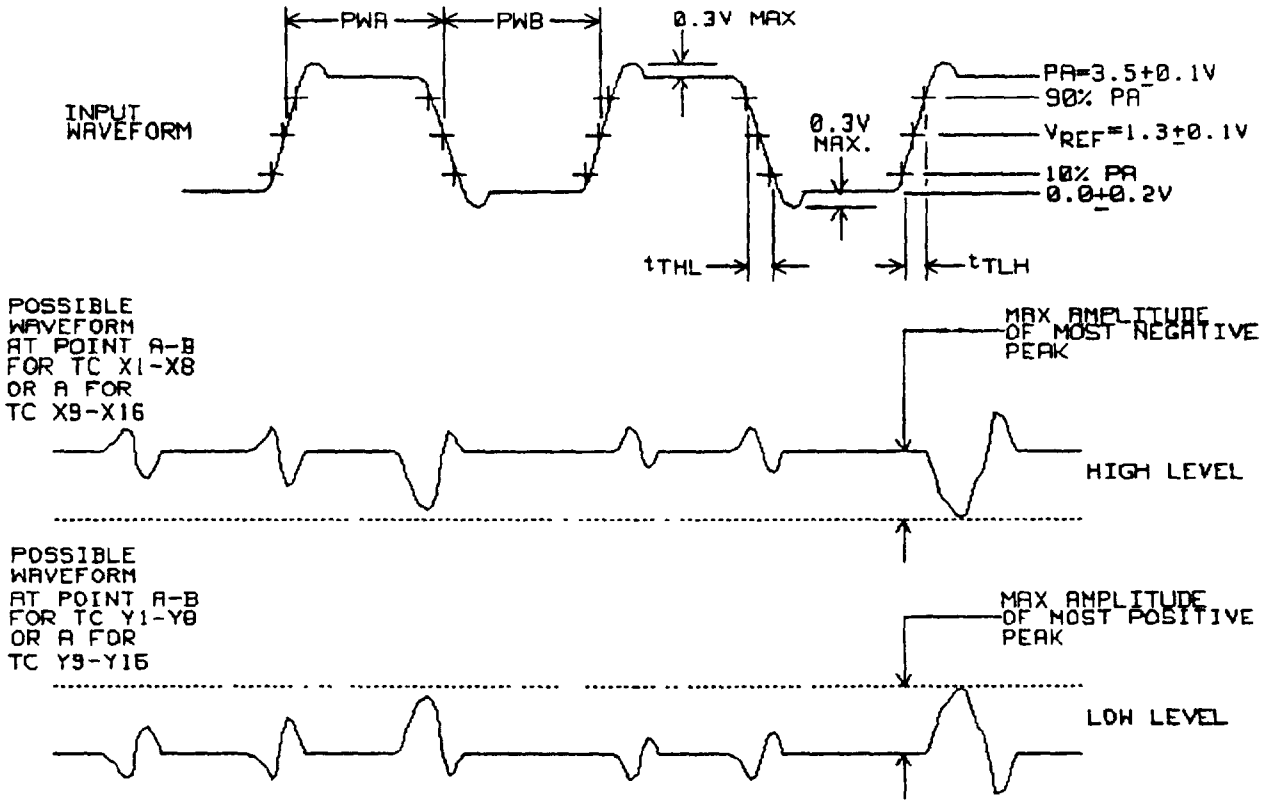


NOTES:

1. R1 = 51.1 Ohms ± 5 percent, 1 W.
2. Load each output as specified on figure 13.

FIGURE 22. Output crosstalk test circuit for tables XXII and XXIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTE:

1. Input parameters: PWA = PWB = 250 nS ± 10 percent, tTHL = tTLH = 3.5 ± 1.5 nS (verified at each input).

FIGURE 23. Output crosstalk test waveforms for tables XXII and XXIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Figs
			Min	Max		
Z	PSTA	1,31	0	500	mV	24,25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

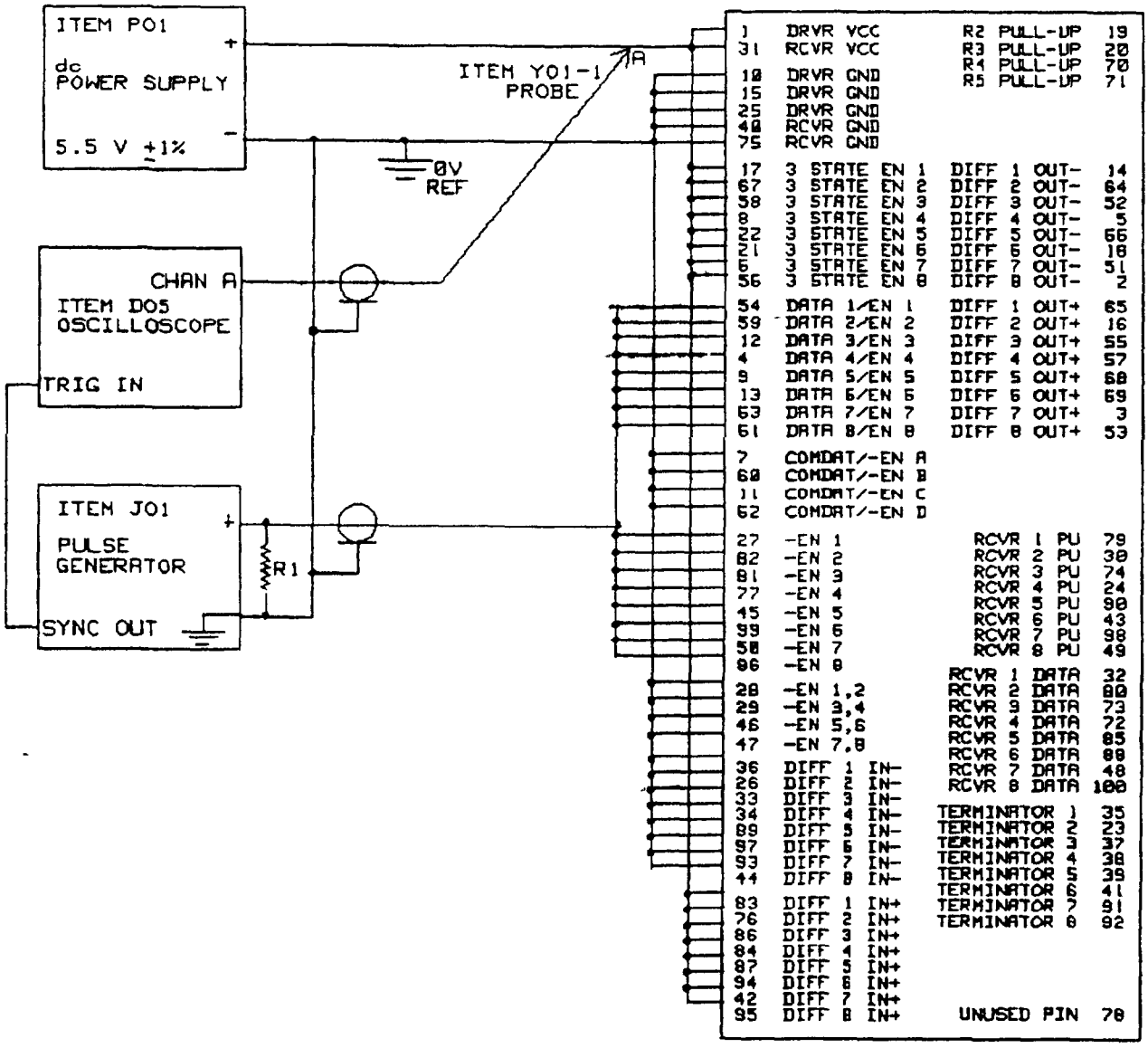
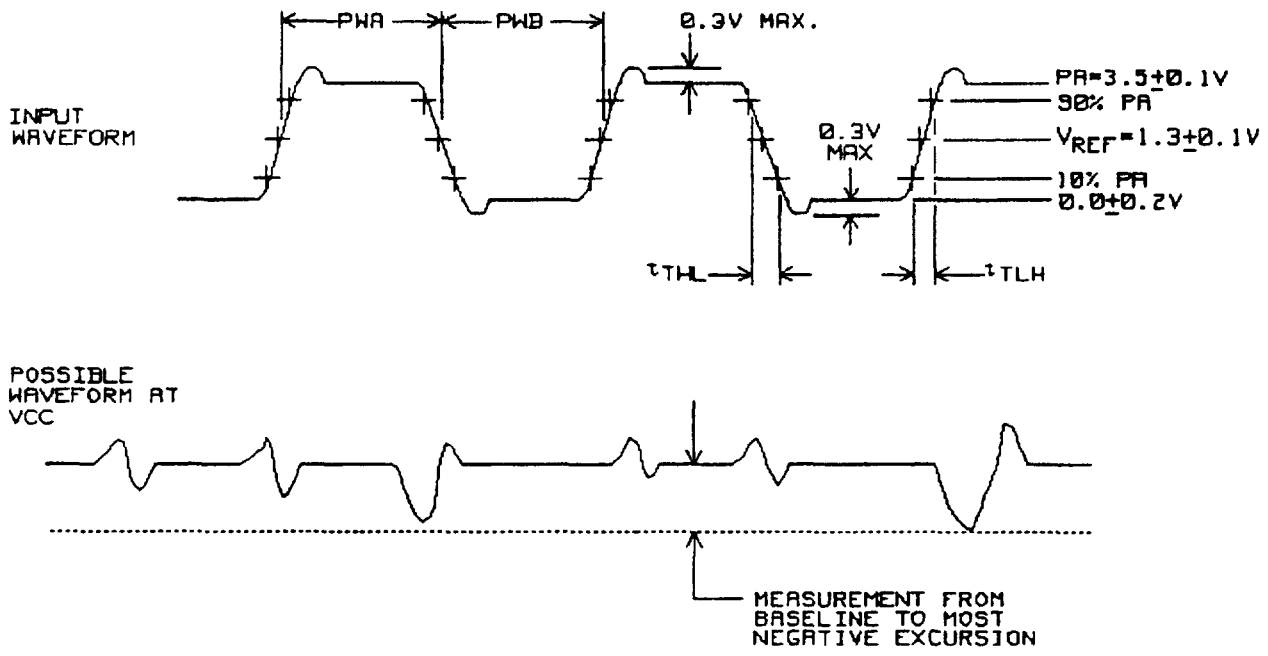


FIGURE 24. Power supply transient amplitude test circuit for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input parameters: PWA = PWB = 250 nS ± 10 percent, tTHL = tTLH = 10 ± 1 nS.

2. Connect point A as close as possible to pins 1 and 31.

FIGURE 25. Power supply transient amplitude waveform for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXV. Power supply filter capacitance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
AA1	PSFC	Open	1	3.6	6.6	uF	26
AA2	PSFC	Open	31	3.6	6.6	uF	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Input capacitance test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
BB1	3 STATE EN 1	17	Open	2	15	pF	26
BB2	3 STATE EN 2	67					
BB3	3 STATE EN 3	58					
BB4	3 STATE EN 4	8					
BB5	3 STATE EN 5	22					
BB6	3 STATE EN 6	21					
BB7	3 STATE EN 7	6					
BB8	3 STATE EN 8	56					
BB9	DATA 1/EN 1	54					
BB10	DATA 2/EN 2	59					
BB11	DATA 3/EN 3	12					
BB12	DATA 4/EN 4	4					
BB13	DATA 5/EN 5	9					
BB14	DATA 6/EN 6	13					
BB15	DATA 7/EN 7	63					
BB16	DATA 8/EN 8	61					
BB17	COMDAT/-EN A	7					
BB18	COMDAT/-EN B	60					
BB19	COMDAT/-EN C	11					
BB20	COMDAT/-EN D	62	Open	2	15	pF	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Input capacitance test - Continued.

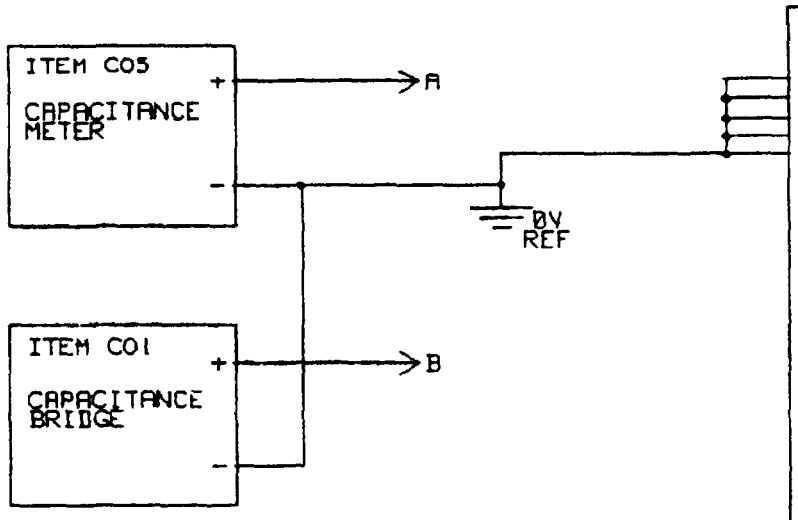
Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
BB21	-EN 1	27	Open	2	15	pF	26
BB22	-EN 2	82					
BB23	-EN 3	81					
BB24	-EN 4	77					
BB25	-EN 5	45					
BB26	-EN 6	99					
BB27	-EN 7	50					
BB28	-EN 8	96					
BB29	-EN 1,2	28					
BB30	-EN 3,4	29					
BB31	-EN 5,6	46					
BB32	-EN 7,8	47					
BB33	DIFF 1 IN-	36					
BB34	DIFF 2 IN-	26					
BB35	DIFF 3 IN-	33					
BB36	DIFF 4 IN-	34					
BB37	DIFF 5 IN-	89					
BB38	DIFF 6 IN-	97					
BB39	DIFF 7 IN-	93					
BB40	DIFF 8 IN-	44	Open	2	15	pF	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Input capacitance test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
BB41	DIFF 1 IN+	83	Open	2	15	pF	26
BB42	DIFF 2 IN+	76					
BB43	DIFF 3 IN+	86					
BB44	DIFF 4 IN+	84					
BB45	DIFF 5 IN+	87					
BB46	DIFF 6 IN+	94					
BB47	DIFF 7 IN+	42					
BB48	DIFF 8 IN+	95	Open	2	15	pF	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



1	DRVR VCC	R2 PULL-UP	19
31	RCVR VCC	R3 PULL-UP	28
		R4 PULL-UP	70
10	DRVR GND	R5 PULL-UP	71
15	DRVR GND		
25	DRVR GND		
40	RCVR GND		
75	RCVR GND		
17	3 STATE EN 1	DIFF 1 OUT-	14
67	3 STATE EN 2	DIFF 2 OUT-	64
58	3 STATE EN 3	DIFF 3 OUT-	52
8	3 STATE EN 4	DIFF 4 OUT-	5
22	3 STATE EN 5	DIFF 5 OUT-	56
21	3 STATE EN 6	DIFF 6 OUT-	18
6	3 STATE EN 7	DIFF 7 OUT-	51
56	3 STATE EN 8	DIFF 8 OUT-	2
54	DATA 1/EN 1	DIFF 1 OUT+	65
59	DATA 2/EN 2	DIFF 2 OUT+	16
12	DATA 3/EN 3	DIFF 3 OUT+	55
4	DATA 4/EN 4	DIFF 4 OUT+	57
9	DATA 5/EN 5	DIFF 5 OUT+	58
13	DATA 6/EN 6	DIFF 6 OUT+	69
63	DATA 7/EN 7	DIFF 7 OUT+	3
61	DATA 8/EN 8	DIFF 8 OUT+	53
7	COMDAT/-EN A		
60	COMDAT/-EN B		
11	COMDAT/-EN C		
62	COMDAT/-EN D		
27	-EN 1	RCVR 1 PU	29
82	-EN 2	RCVR 2 PU	30
81	-EN 3	RCVR 3 PU	24
77	-EN 4	RCVR 4 PU	24
45	-EN 5	RCVR 5 PU	90
88	-EN 6	RCVR 6 PU	43
50	-EN 7	RCVR 7 PU	98
96	-EN 8	RCVR 8 PU	49
28	-EN 1,2	RCVR 1 DATA	32
29	-EN 3,4	RCVR 2 DATA	80
46	-EN 5,6	RCVR 3 DATA	73
47	-EN 7,8	RCVR 4 DATA	72
		RCVR 5 DATA	85
		RCVR 6 DATA	88
36	DIFF 1 IN-	RCVR 7 DATA	48
26	DIFF 2 IN-	RCVR 8 DATA	100
33	DIFF 3 IN-		
34	DIFF 4 IN-	TERMINATOR 1	35
89	DIFF 5 IN-	TERMINATOR 2	23
97	DIFF 6 IN-	TERMINATOR 3	37
93	DIFF 7 IN-	TERMINATOR 4	38
44	DIFF 8 IN-	TERMINATOR 5	39
		TERMINATOR 6	41
83	DIFF 1 IN+	TERMINATOR 7	91
76	DIFF 2 IN+	TERMINATOR 8	92
86	DIFF 3 IN+		
84	DIFF 4 IN+		
87	DIFF 5 IN+		
94	DIFF 6 IN+		
42	DIFF 7 IN+		
95	DIFF 8 IN+	UNUSED PIN	78

Capacitance bridge (C01) setup:
Frequency = 1 KHz
Vpk = 0.3 V max.

NOTES:

1. Null out the effect of the test leads prior to connecting to the module.

2. Do not attach test connectors or wires to any of the module pins other than pins 10, 15, 25, 40, 75 and the pin under test. Stray capacitance added to other pins may affect the readings.

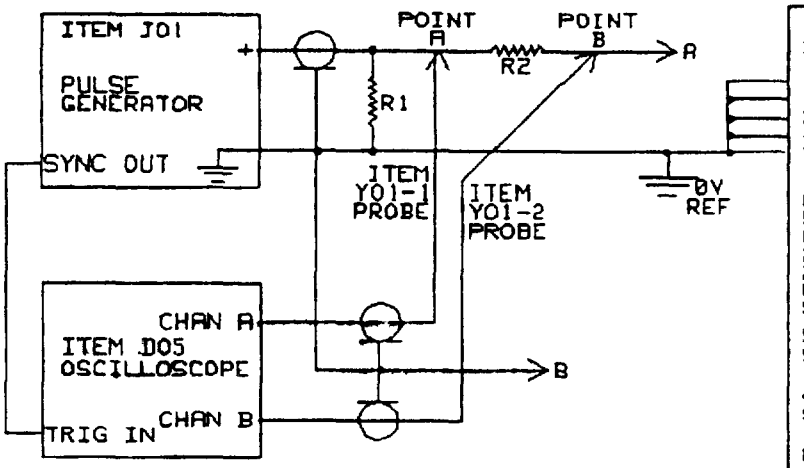
FIGURE 26. Power supply filter and input capacitance test circuit for tables XXV and XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Terminator test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
CC1	TERMINATOR 1	35	83	0.394	0.591	V	27,28
CC2	TERMINATOR 2	23	76				
CC3	TERMINATOR 3	37	86				
CC4	TERMINATOR 4	38	84				
CC5	TERMINATOR 5	39	87				
CC6	TERMINATOR 6	41	94				
CC7	TERMINATOR 7	91	42				
CC8	TERMINATOR 8	92	95	0.394	0.591	V	27,28
CC9	TERMINATOR 1	35	83	2.79	4.18	V	27,28
CC10	TERMINATOR 2	23	76				
CC11	TERMINATOR 3	37	86				
CC12	TERMINATOR 4	38	84				
CC13	TERMINATOR 5	39	87				
CC14	TERMINATOR 6	41	94				
CC15	TERMINATOR 7	91	42				
CC16	TERMINATOR 8	92	95	2.79	4.18	V	27,28

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



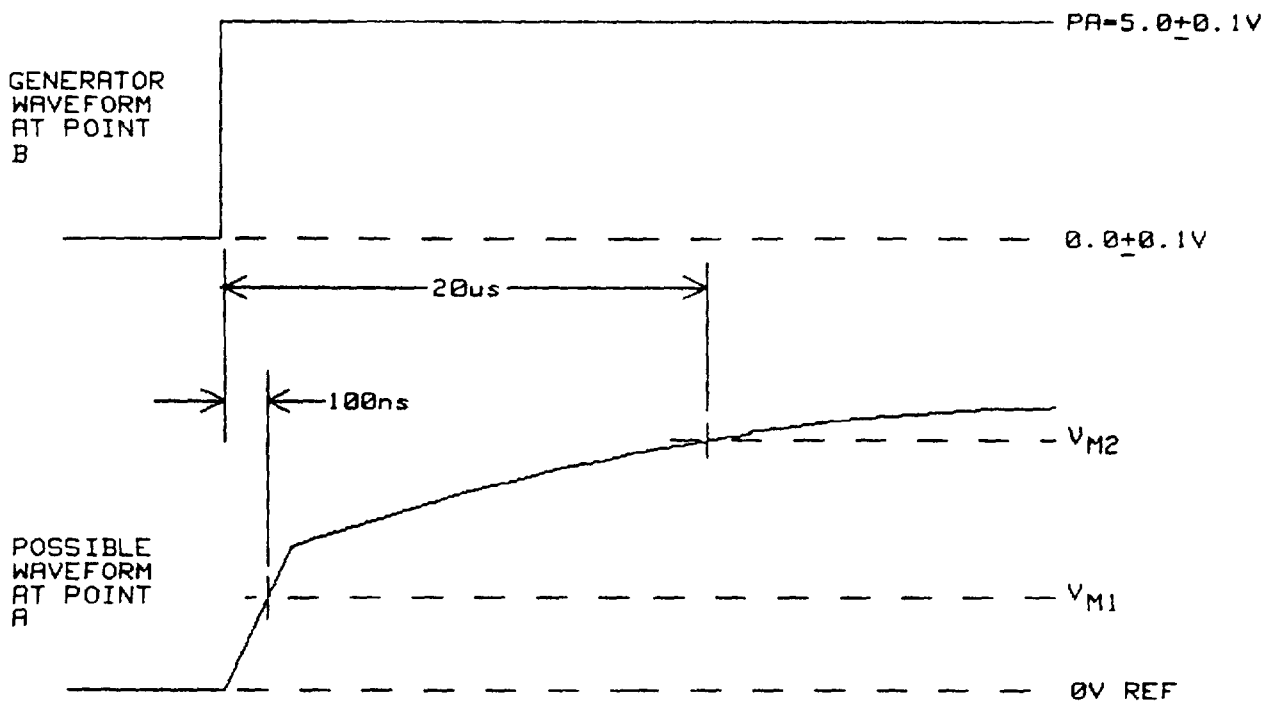
1	DRVR VCC	R2 PULL-UP	19
31	RCVR VCC	R3 PULL-UP	20
		R4 PULL-UP	70
10	DRVR GND	R5 PULL-UP	71
15	DRVR GND		
25	DRVR GND		
40	RCVR GND		
75	RCVR GND		
17	3 STATE EN 1	DIFF 1 OUT-	14
67	3 STATE EN 2	DIFF 2 OUT-	64
58	3 STATE EN 3	DIFF 3 OUT-	52
8	3 STATE EN 4	DIFF 4 OUT-	5
22	3 STATE EN 5	DIFF 5 OUT-	66
21	3 STATE EN 6	DIFF 6 OUT-	10
6	3 STATE EN 7	DIFF 7 OUT-	51
56	3 STATE EN 8	DIFF 8 OUT-	2
54	DATA 1/EN 1	DIFF 1 OUT+	65
59	DATA 2/EN 2	DIFF 2 OUT+	16
12	DATA 3/EN 3	DIFF 3 OUT+	55
4	DATA 4/EN 4	DIFF 4 OUT+	57
9	DATA 5/EN 5	DIFF 5 OUT+	68
13	DATA 6/EN 6	DIFF 6 OUT+	69
53	DATA 7/EN 7	DIFF 7 OUT+	3
61	DATA 8/EN 8	DIFF 8 OUT+	53
7	COMDAT/-EN A		
60	COMDAT/-EN B		
11	COMDAT/-EN C		
62	COMDAT/-EN D		
27	-EN 1	RCVR 1 PU	79
82	-EN 2	RCVR 2 PU	30
81	-EN 3	RCVR 3 PU	74
77	-EN 4	RCVR 4 PU	24
45	-EN 5	RCVR 5 PU	90
99	-EN 6	RCVR 6 PU	43
50	-EN 7	RCVR 7 PU	98
96	-EN 8	RCVR 8 PU	49
		RCVR 1 DATA	32
28	-EN 1,2	RCVR 2 DATA	80
29	-EN 3,4	RCVR 3 DATA	73
46	-EN 5,6	RCVR 4 DATA	72
47	-EN 7,8	RCVR 5 DATA	85
		RCVR 6 DATA	88
36	DIFF 1 IN-	RCVR 7 DATA	48
26	DIFF 2 IN-	RCVR 8 DATA	100
33	DIFF 3 IN-		
34	DIFF 4 IN-	TERMINATOR 1	35
89	DIFF 5 IN-	TERMINATOR 2	23
97	DIFF 6 IN-	TERMINATOR 3	37
93	DIFF 7 IN-	TERMINATOR 4	38
44	DIFF 8 IN-	TERMINATOR 5	39
		TERMINATOR 6	41
83	DIFF 1 IN+	TERMINATOR 7	91
76	DIFF 2 IN+	TERMINATOR 8	92
86	DIFF 3 IN+		
84	DIFF 4 IN+		
87	DIFF 5 IN+		
94	DIFF 6 IN+		
42	DIFF 7 IN+		
95	DIFF 8 IN+	UNUSED PIN	78

NOTE:

1. R1 = 51.1 Ohms ± 5 percent, 1 W; R2 = 750 Ohms ± 1/8 W.

FIGURE 27. Terminator test circuit for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: PW = 25 us ± 10 percent, PRR = 8 kHz ± 10 percent, tTLH = tTHL = 10 ± 2 nS (referenced at the 10 percent and 90 percent PA points).

2. Measure the voltage from Vref at the 100 nS reference (Vm1, test conditions CC1-CC8) and the 20 nS reference (Vm2, test conditions CC9-CC16).

3. Timing references shall be from the beginning of the input pulse (generator pulse at point B).

FIGURE 28. Terminator test circuit waveforms for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7.

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

1. RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-W-28787/362	2 DOCUMENT DATE (YYMMDD) 1990 October 26

3. DOCUMENT TITLE
Modules, Digital, Standard Electronic Driver/Receiver Differential Module, Key Code HJN

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A)	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-1528	(2) AUTOVON 88-482-1528
c ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F Dilger, Bldg 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041 3466 Telephone (703) 756 2340 AUTOVON 289-2340	

LEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:05Z
Check the source to verify that this is the current version before use.